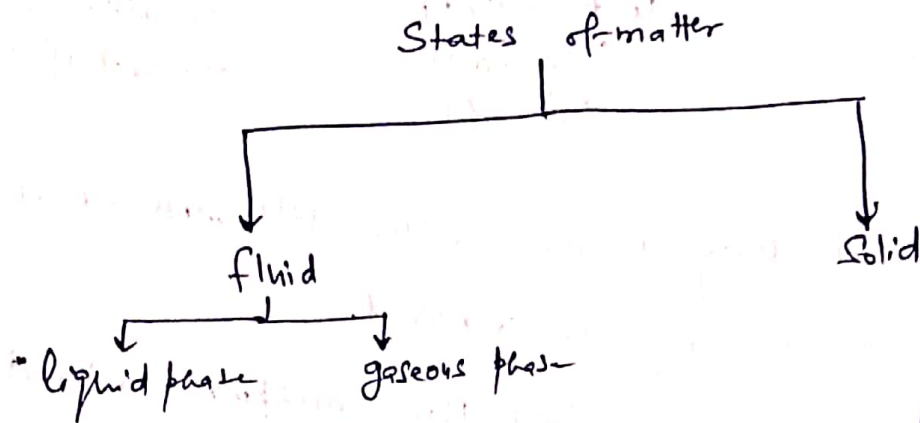


Basic definition and fluid properties

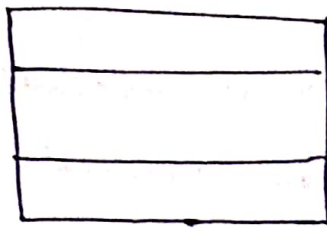
- All substances consists of vast number of molecules separated by empty space. Based on intermolecular attraction forces states of matter are classified as



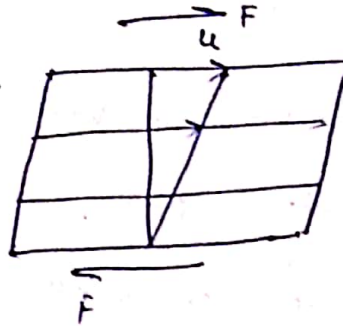
- Solids have ^{very} strong intermolecular attraction forces and hence they are rigid
- Liquids have relatively weaker intermolecular attraction forces but sufficient to keep the liquid together in a definite volume
- In gases intermolecular attraction forces are extremely weak and gas fills the container in which it is placed.

Definition of fluid

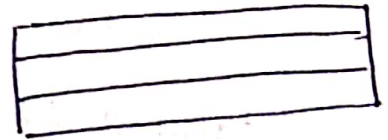
- " A fluid is a substance which deforms continuously when subjected to shear forces, however small the shear force may be."
- The act of continuous deformation is called the flow.
- Hence a fluid will flow when subjected to shear forces.



Initial state
(rest)



Under action of
Shear forces
(flowing)



"no tendency to regain
its initial state on
removal of forces"

- There is no tendency to regain initial state, instead it comes to rest as soon as forces are removed.
- The state of rest of a fluid thus corresponds to the absence of shear forces.
- Solid suffers strain when subjected to shear forces, whereas fluid suffers a rate of strain i.e., flows.

Concept of Continuum

- Although fluid consists of discrete molecules, the fluid is treated as continuous medium. All the voids and spaces which may occur in the fluids are ignored. The physical properties of matter can be considered as continuous function of space and time. The assumption of matter as continuous medium is called continuum e.g., Solid Continuum and fluid Continuum. Hence study of fluids in Continuum fluid mechanics ignores molecular structure.

Fluid element or particle

- Infinitesimal lump of fluid is called a fluid element or a particle.
- It is assumed to have very small but nonzero dimensions.
- It may be chosen to at any point and its shape is arbitrary eg. triangular prism or rectangular parallelepiped.
- Particle should not be confused with atoms or molecules.

Control volume

- An identified region fixed in space.
- It is bounded by control surface.
- An infinitesimal control volume is called differential control volume.
- Net rate of change of fluid mass, momentum and energy across control surface are called mass flux, momentum flux and energy flux, respectively.

Forces

- External action on a body which tends to change state, orientation, shape and size of the body.

POINT FORCE: Acting on very small but nonzero elemental area.

LINE FORCE: These are assumed to act over a line element eg. surface tension force.

SURFACE FORCE: Force distributed over surface area e.g. for a fluid mass force due to pressure and shearing action.

BODY FORCE: Force distributed over volume of body. eg. gravitational force (weight).

Properties of fluids

- Certain characteristics of a continuous fluid are independent of the motion of fluid called basic properties.

DENSITY : It is defined as mass of substance per unit volume.

If a fluid element enclosing a point P has volume δV and mass δm . Then density at that point P.

$$\rho = \lim_{\delta V \rightarrow 0} \frac{\delta m}{\delta V} = \frac{dm}{dV} \quad \text{Kg/m}^3$$

- Mass density is a local fluid property which varies from point to point as well as at a point with passage of time.
- Density $\propto$ no. of molecules per unit volume.
as $T \uparrow$, $\rho \downarrow$ (Because less no. of molecules exist in given volume as $T \uparrow$)
as $P \uparrow$, $\rho \uparrow$ (Because large no. of molecules are forced in a given volume)

SPECIFIC WEIGHT

Weight of a substance per unit volume is called specific weight

$$\gamma = \lim_{\delta V \rightarrow 0} \frac{\delta m \cdot g}{\delta V} = g \frac{dm}{dV} = \rho g \quad \frac{\text{N}}{\text{m}^3}$$

SPECIFIC VOLUME

It is defined as volume per unit mass of fluid.

$$v = \frac{1}{\rho} = \frac{\text{m}^3}{\text{kg}}$$

Facts!

- For liquids ρ , γ and ν varies slightly with P and T variation because molecules are arranged compactly as compared to gases.
- For gases they vary greatly with P and T variations.
- Presence of dissolved air, salts in solution and suspended matter will slightly increase the value of density.

SPECIFIC GRAVITY (S)

- Specific gravity of a liquid is the ratio of its density to that of water at 4°C .
 - Specific gravity of a gas is the ratio of its density to that of air.
- "Water and air are being accepted as reference liquid and gaseous fluids, respectively."

Incompressible fluid

"A fluid is said to be incompressible if the density of fluid does not change (or changes very little) with a change of pressure or temperature or velocity.
(Liquids are incompressible)"

Compressible fluid

"A fluid is said to be compressible if its density changes with change in pressure or temperature or velocity."
(Gaseous are compressible)"

Ideal fluid

- Ideal fluid is a fluid which is incompressible, have no viscosity (i.e. frictionless) and no surface tension.
- There is no resistance to flow of such fluids.
- Ideal fluids are imaginary ^{and no such fluids exist in nature} but fluids having very low viscosity such as water and air can be treated as ideal fluids for practical purpose.
- Since water is incompressible, it is more near to ideal fluid than air.

Real fluid

- They are practical fluids available in nature.
- These fluids possess property such as viscosity, surface tension, and compressibility.
- They always offer resistance to flow.

Viscosity

- "Viscosity can be defined as that property of fluid which resist relative motion of its adjacent layers".
- It is a measure of internal fluid friction which cause resistance to flow.
- The property of viscosity is present in all real fluids and it distinguishes them from non viscous ideal fluids.
- molasses, tar and glycerine are example of highly viscous liquids.
- Water, air and petrol have very low viscosity and hence they flow much more easily and rapidly.

Newton's law of Viscosity

- Application of shear force (shear stress) produces rate of strain in a fluid i.e. flow.
- Newton related shear stress with the rate of strain at a point in a fluid
- "According to Newton's law of viscosity, the shear stress on a fluid element layer is directly proportional to rate of strain, the constant of proportionality being called coefficient of viscosity."

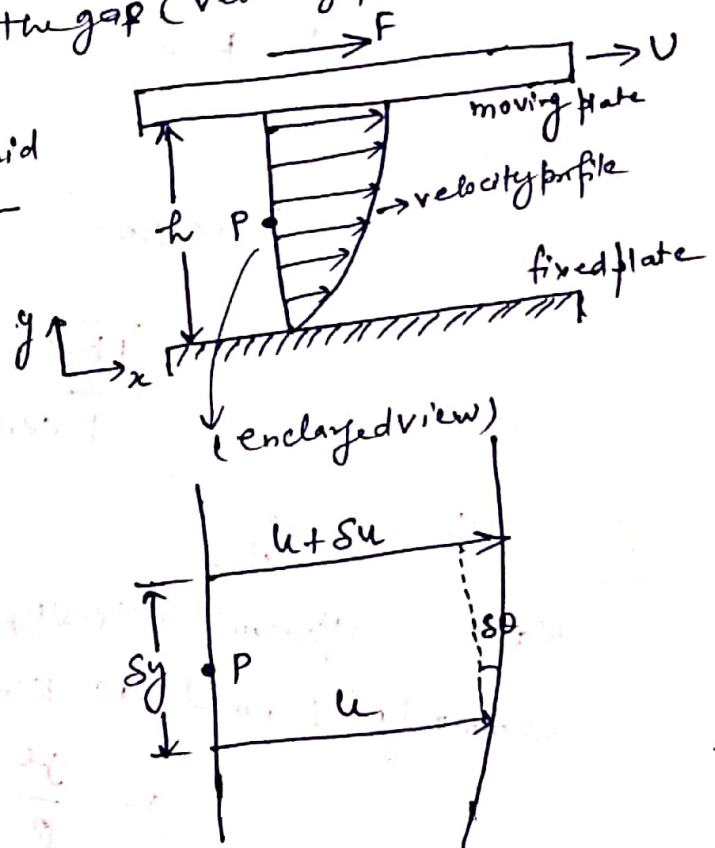
Consider fluid contained between two parallel plates distance h apart. Lower plate is held stationary and upper plate is moved at a constant velocity U by application of force F . The fluid will be subjected to shearing force (shearing stress) and velocity will vary across the gap (velocity profile shown).

- Rate of strain at a point in fluid is the rate of shearing of adjacent layers at that point.
- i.e. rate of change of angle of the line joining the fluid layers just above and below point P.

$$\begin{aligned} \theta &\approx \tan \theta = \frac{(u + \delta u) \delta t - u \delta t}{\delta y} \\ &= \frac{\delta u}{\delta y} \delta t \end{aligned}$$

$$\text{Rate of strain} = \lim_{\delta t \rightarrow 0} \left(\frac{\delta u}{\delta y} \frac{\delta t}{\delta t} \right)$$

$$\boxed{\epsilon = \frac{du}{dy}}$$



(δt = small time gap)

According to Newton's law

Shear stress $\propto$ rate of strain

$$\tau \propto \epsilon$$

$$\tau \propto \frac{du}{dy}$$

$$\tau = \eta \frac{du}{dy}$$

$$\tau = F/A$$

A = Surface area of plate

→ Newton's law of viscosity

η = Coefficient of viscosity, also called absolute viscosity or dynamic viscosity.

SI units : $\frac{N \cdot s}{m^2}$ or Pa-s or $\frac{kg}{m \cdot s}$

C.G.S. units : Unit of viscosity is poise named after the scientist Poiseuille.

$$1 \text{ poise} = \frac{1 \text{ dyne} \cdot s}{cm^2}$$

$$1 N = 10^5 \text{ dyne}$$

$$1 \text{ poise} = \frac{10^5 \text{ N} \cdot s}{10^{-4} m^2} = 0.1 \text{ Pa} \cdot s$$

$$1 \text{ poise} = 0.1 \text{ Pa} \cdot s$$

$$1 \text{ centipoise} = 10^{-3} \text{ Pa} \cdot s$$

Kinematic viscosity

Ratio of dynamic viscosity to the fluid density is commonly termed as kinematic viscosity (ν).

$$\nu = \frac{\eta}{\rho} = \frac{kg/m \cdot s}{kg/m^3} = \frac{m^2}{s}$$

$$C.G.S. \text{ unit: } 1 \text{ stoke} = 1 \frac{cm^2}{s} = 10^{-4} \frac{m^2}{s}$$

$$1 \text{ centistoke} = 10^{-6} \frac{m^2}{s}$$

— Since ν involves only the magnitude of length and time hence called kinematic viscosity.

Effect of temperature and pressure on viscosity

- The origin of viscosity is due to cohesion and molecular momentum transfer between fluid layers.

Note: Cohesion means intermolecular attraction forces between the molecules of the same liquid.

- Viscosity of liquids drops with increase in temperature whereas reverse happens for gases.
- In liquids, viscosity is governed by cohesive forces due to closely packed molecules and as $T \uparrow$, these forces decrease and viscosity decreases.
- In gases, cohesive forces are small and molecular momentum transfer predominates. As $T \uparrow$ molecular activity increases and causes an increase in viscosity.
- Viscosity of liquid or gas is practically independent of pressure for the range that is ordinarily encountered in practice.

Classification of fluids

Newtonian and Non-Newtonian fluids

fluids which obey Newton's linear law of viscosity

$$\tau = \mu \frac{dy}{dx}$$

are called Newtonian fluids.

fluids which does not obey Newton's linear law of viscosity are called Non-Newtonian fluids

$$\tau \neq \mu \frac{dy}{dx}$$

Tutorial problem (Solved)

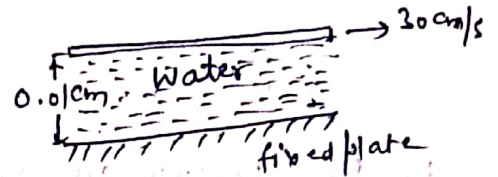
A flat plate 0.1 m^2 area is pulled at 30 cm/s relative to another plate located at a distance of 0.01 cm from it. The fluid separating between them being water with $\mu = 0.001 \text{ Ns/m}^2$. Find the force and power required to maintain the velocity.

Solution

$$du = \frac{30}{100} = 0.3 \text{ m/s}$$

$$dy = 0.01 \times 10^{-2} \text{ m}$$

$$\tau = \frac{\mu dy}{dy} = 0.001 \times \frac{0.3}{0.01 \times 10^{-2}} = 3 \text{ N/m}^2$$



$$\text{Force required} = \tau \times A = 3 \times 0.1 = 0.3 \text{ Newtons}$$

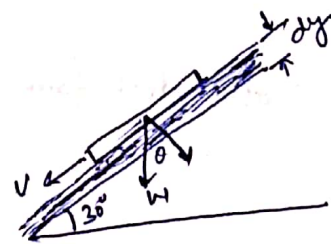
$$\text{Power required} = F \times V = 0.3 \times 0.3 = 0.09 \text{ Watts}$$

Tutorial problem (Solved)

A 1 m^2 , 10 kg flat plate slides down 30° inclined plane over 0.1 cm uniform layer of oil, $\mu = 0.05 \frac{\text{Ns}}{\text{m}^2}$, determine steady state velocity.

Solution

$U =$ Steady state velocity of plate on inclined plane moving over uniform layer of oil



$$\text{Viscous resistance} = \tau \times A = \frac{\mu du}{dy} \times A$$

This force is balanced by weight of plate along slope $= W \sin \theta$

$$0.05 \times \frac{(U-0)}{0.1 \times 10^{-2}} \times 1 = 10 \times 9.8 \times \sin 30$$

$$U = 0.98 \text{ m/s}$$

Tutorial problem (Solved)

Through a narrow gap of height h a thin plate of large extent is pulled at a velocity V . on one side of plate is oil of viscosity μ_1 and on another side oil of viscosity μ_2 . Calculate position of plate so that

- (i) Shear force on two sides of plate is equal
- (ii) pull required to drag the plate is minimum.

Solution

- (i) Force per unit area on upper surface of plate

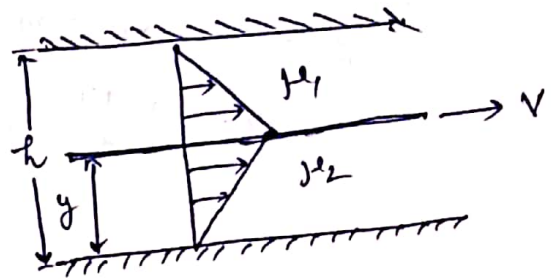
$$F_1 = \mu_1 \frac{dy}{dy} = \mu_1 \frac{V}{h-y}$$

Force per unit area on lower surface of plate

$$F_2 = \mu_2 \frac{dy}{dy} = \mu_2 \frac{V}{y}$$

$$F_1 = F_2 \Rightarrow$$

$$\Rightarrow \boxed{y = \frac{\mu_2 h}{\mu_1 + \mu_2}}$$



- (ii) Total drag force required to pull the plate

$$F = F_1 + F_2$$

$$= \mu_1 \frac{V}{h-y} + \mu_2 \frac{V}{y}$$

For F to be minimum

$$\frac{dF}{dy} = 0 \Rightarrow -\mu_1 \frac{V(-1)}{(h-y)^2} - \mu_2 \frac{V}{y^2}$$

$$(h-y)^2 \mu_2 = \mu_1 y^2$$

$$\left(\frac{h}{y} - 1\right)^2 = \frac{\mu_1}{\mu_2}$$

$$\boxed{y = \frac{h}{1 + \sqrt{\mu_1/\mu_2}}}$$

FLUID STATICS

State of rest and Static pressure

- State of absolute rest refers to zero velocity of each elements of fluid.
 - There is no relative velocity between fluid elements i.e. no rate of strain and no shear stresses.
- " When a fluid is contained in a vessel, It's having state of rest because there is no force component tangential to walls of container but fluid exerts normal forces on the sides and bottom of the container"
- This normal force per unit area is called "Static pressure".

mathematically

Pressure at a point in fluid at rest is

$$P = \lim_{\Delta A \rightarrow 0} \frac{\Delta F}{\Delta A}$$

ΔF = force acting normal to small area ΔA

Units

SI

N/m^2

or

Pa

(1 bar = $10^5 N/m^2$)

mks

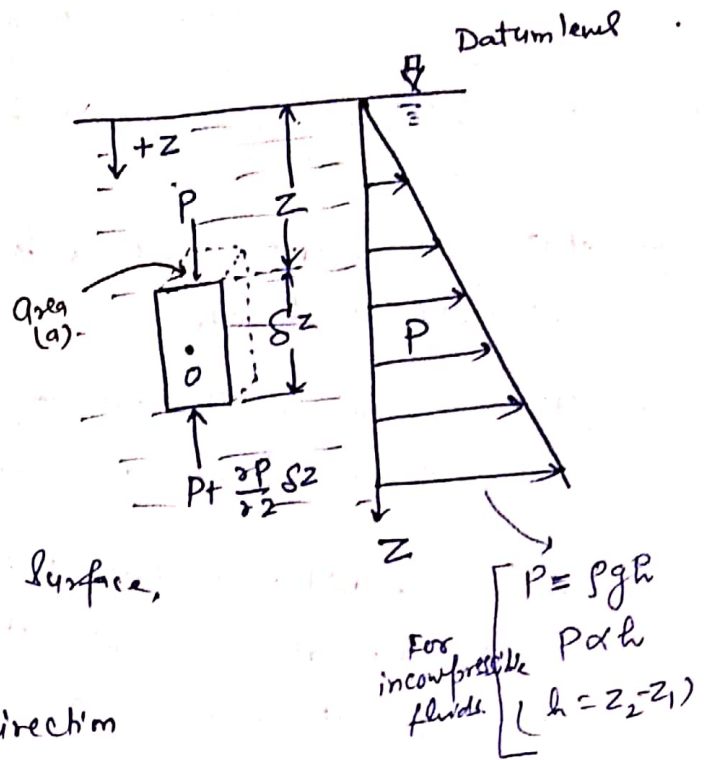
$\frac{kgf}{cm^2}$

or

$\frac{kgf}{cm^2}$

Hydrostatic law

- To find pressure at a point O in the fluid at rest
- Enclose the point by vertical fluid prism of cross section area a and a length δz .



- At a depth z below the free surface, the pressure is P

For equilibrium in z -direction

Net pressure force acting upwards

$$(P + \frac{\partial P}{\partial z} \delta z) a - Pa = \frac{\partial P}{\partial z} \delta z a$$

Gravity force on elementary prism equals the weight of prism
 $= \rho g (\delta z \cdot a)$

for equilibrium of prism in z -direction

$$\frac{\partial P}{\partial z} \delta z a = \rho g \delta z a$$

$$\boxed{\frac{\partial P}{\partial z} = \rho g} \rightarrow \text{Hydrostatic law}$$

"Rate of increase of pressure in vertically downward direction at a point in static fluid must be equal to local specific weight of fluid"

- For equilibrium in x and y -directions

$$\frac{\partial P}{\partial x} = 0 \quad \text{and} \quad \frac{\partial P}{\partial y} = 0$$

hence we can write $\boxed{\frac{\partial P}{\partial z} = \rho g}$ - Hydrostatic law

- This law is valid for incompressible and compressible fluids as local density ρ is taken into account.
- The law is valid for viscous or inviscid fluids.

INCOMPRESSIBLE FLUIDS

$$\rho = \text{constant}, \quad \int dp = \int \rho g dz$$

$$P_2 - P_1 = \rho g (z_2 - z_1) \Rightarrow \rho g h$$

$$: \quad h = z_2 - z_1, \quad \text{if } P_1 = P_{\text{atm}} \\ P_2 - P_{\text{atm}} = \rho g h$$

$$\boxed{P = \rho g h}$$

linear varying - $\boxed{P \propto h}$

Triangular distribution of pressure for incompressible fluids

UNITS of pressure

$$1 \text{ standard atm} = 1.01325 \text{ bar} = 1.01325 \times 10^5 \text{ N/m}^2$$

$$\text{In m of WG} = \frac{P}{\rho g} = \frac{1.01325 \times 10^5}{1000 \times 9.81} = 10.33 \text{ m WG}$$

$$\text{In m of HG} = \frac{10.33}{13.6} = 0.76 \text{ m HG} = 760 \text{ mm HG}$$

Pascal's law

"It states that pressure at a point in a static fluid has the same magnitude in all directions"

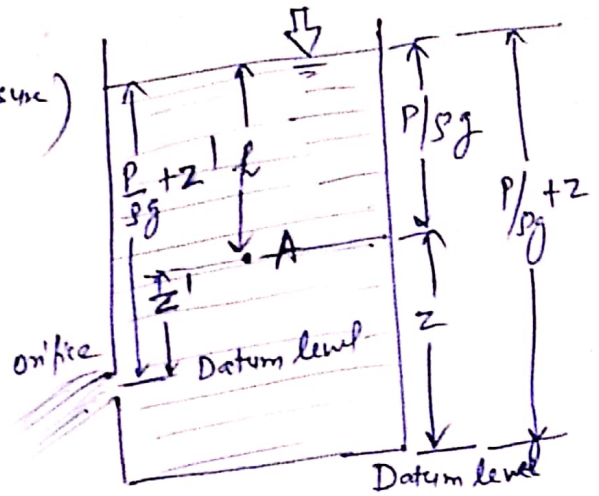
Piezometric head

PRESSURE HEAD

Vertical height of free surface above any point in a liquid at rest is known as pressure head.
i.e. height of the column of that liquid.

$$h = \frac{P}{\rho g}$$

(P is the static pressure whether fluid is in rest or motion)



- the point at which pressure is being considered may have a datum head 'z' with reference to an arbitrary datum level.

PIEZOMETRIC HEAD

"Piezometric head at any point in a fluid at rest or in motion is defined as sum of pressure head and datum head at that point with reference to selected datum level."

$$\text{Piezometric head} = \frac{P}{\rho g} + z$$

$$= \text{Pressure head} + \text{datum head}$$

- Piezometric head at point A with respect to to base of container as datum level and orifice level as datum are shown.
- Sum of pressure head and datum head remains ^{constant} at all points in fluid when referred to a datum level.

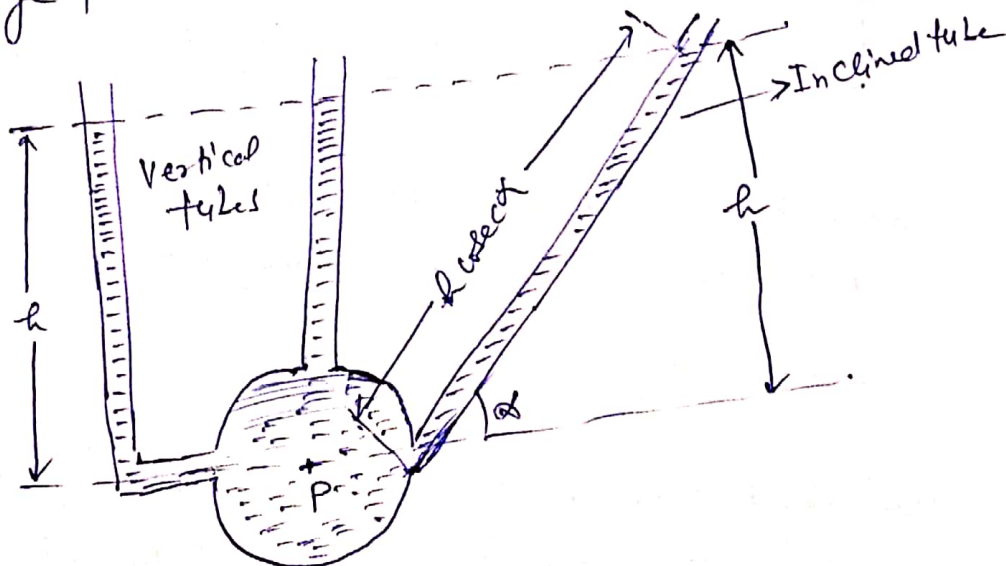
Manometry

"The technique of pressure measurement using hydrostatic law $\frac{dp}{dz} = \rho g$ is known as manometry and the devices used for measurement are called manometers."

- The intensity of fluid pressure P is measured by balancing against a hydrostatic column of liquid of known specific weight.
- Choice of manometric liquids depends upon the range of pressure to be measured and sensitivity required.

Single tube manometers (Piezometer)

- A single vertical tube is used to measure pressure at a point in a liquid.
- lower end of tube is attached to the side of vessel containing liquid and upper end of tube is open to atmosphere. If the pressure is higher than the local atmospheric pressure, liquid will rise in tube.
- Gauge pressure at lower end of tube, $P = \rho g h$



Some facts about piezometer

- It is used to measure pressure if it is above atmospheric pressure, then the liquid will rise in tube.
- Tube can not be used to measure vacuum because in that case the air from atmosphere will be drawn into liquid.
- Tube is also inadequate for very small and very large gage pressures. Because it will require very lengthy tube or height h is too small to be measured accurately.
- For small gage pressures, P , it is better to use an inclined single tube. Reading l recorded on scale is higher than measured by vertical tube by a factor $\csc \alpha$.

$$l = h \csc \alpha$$

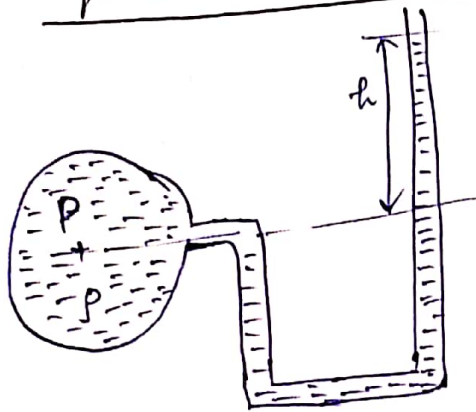
α = inclination from horizontal.

- as α is small, l is increased but lower limit of α is 20° for water in glass tube manometers. This limit is provided by surface tension and capillary effects for accurate measurement.
- Piezometer can be used to measure pressure in liquids only. It can not be used to measure pressure in a gas.
- In order to prevent capillary effect the diameter of piezometer tube should not be less than 10 mm.

Double Column or U-tube manometer

- A U-tube manometer consists of two tubes joined at ends to form U-shaped tube.
- This can be used to measure pressure in liquid or gas.
- Pressure above and below atmospheric can be measured comfortably.
- U-tube can be used as Upright, inclined or inverted.

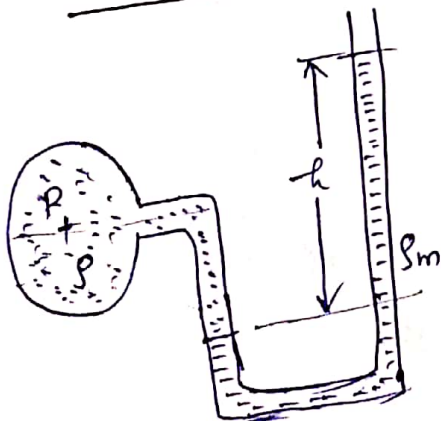
Liquids under moderate pressure



level of liquid rises in open ended tube above the point

$$P = \rho g h$$

Gas under pressure

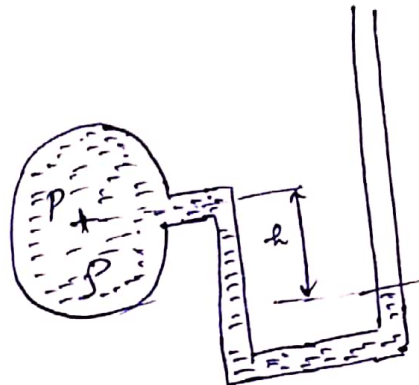


$$P = \rho_m g h, \text{ since } P \ll \rho_m$$

(Column of gas is neglected)

- Use manometric liquid of density ρ_m is selected so that it does not dissolve the gases and the range of pressure rise h of the level of liquid is reasonable and measurable.

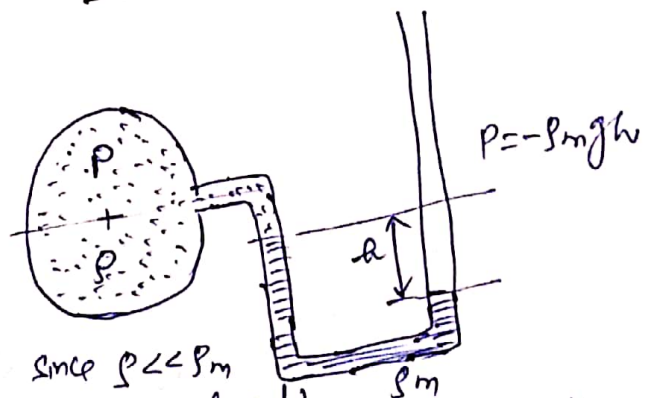
Liquid under Vacuum



level of liquid falls a depression h below the point.

$$P = -\rho g h$$

Gas under Vacuum

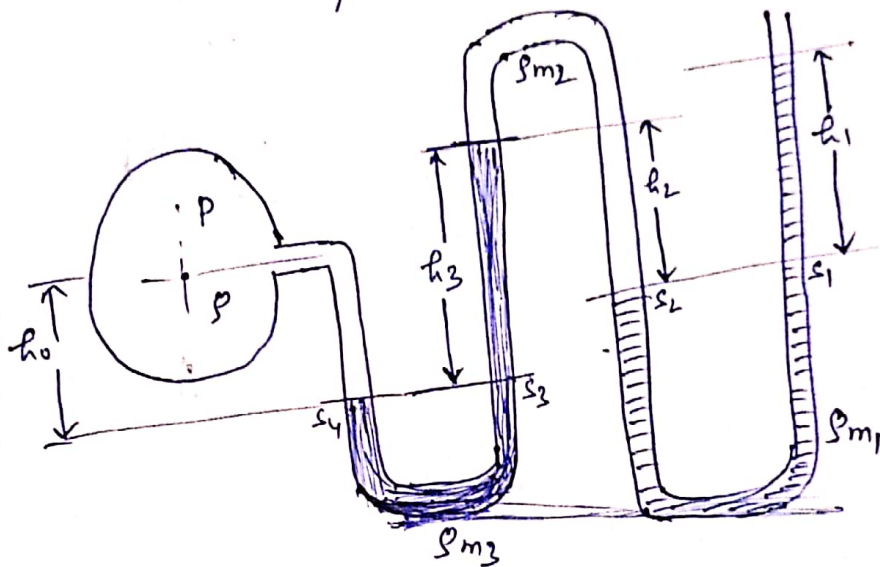


$$P = -\rho_m g h$$

- Water, Alcohol or mercury may be chosen as manometric liquids.
- Alcohol - Low pressure air ($\sim 0.2 \text{ kN/m}^2$)
- Water - moderate pressure air ($\sim 2 \text{ kN/m}^2$)
- mercury - High pressure air ($\sim 20 \text{ kN/m}^2$)

For a fluid under very High pressure

- Here single U-tube will not serve the purpose as the height may be too tall.
- A number of U-tubes of reasonable size may be used with different manometric fluids or same manometric fluid.



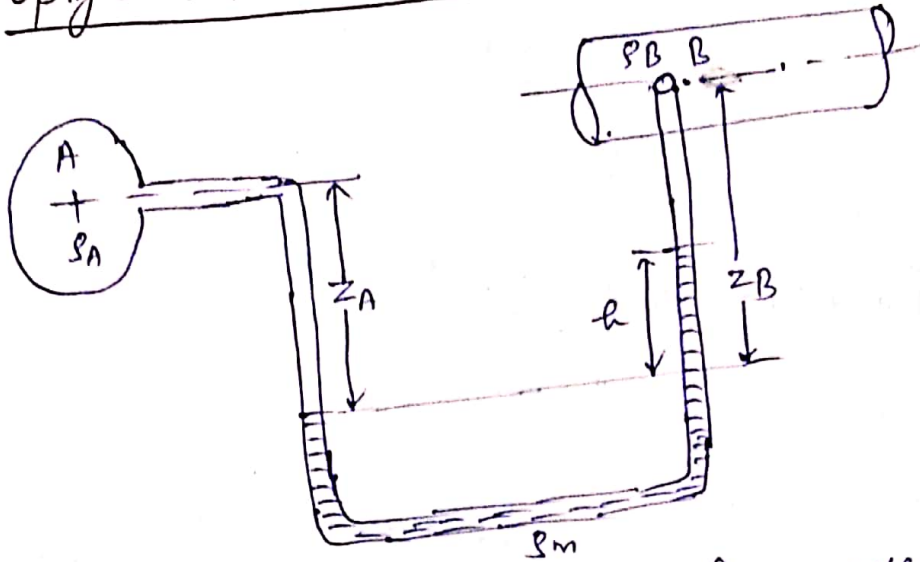
$$P + \rho g h_0 - \rho_{m3} g h_3 + \rho_{m2} g h_2 - \rho_{m1} g h_1 = 0$$

$$P = \rho_{m1} g h_1 - \rho_{m2} g h_2 + \rho_{m3} g h_3 - \rho g h_0$$

Differential manometers

- A differential manometer is used to measure the difference in pressure between two points.
- The two points may either be located in the same vessel or in different vessels containing same or different liquids.

⑨ Upright U-tube manometer



- Two points A and B are in different vessels at different datum levels and different fluids.

$$P_A + S_A g z_A - S_m g h - P_B g (z_B - h) = P_B$$

$$P_A - P_B = S_m g h + P_B g (z_B - h) - S_A g z_A$$

If fluids are same then

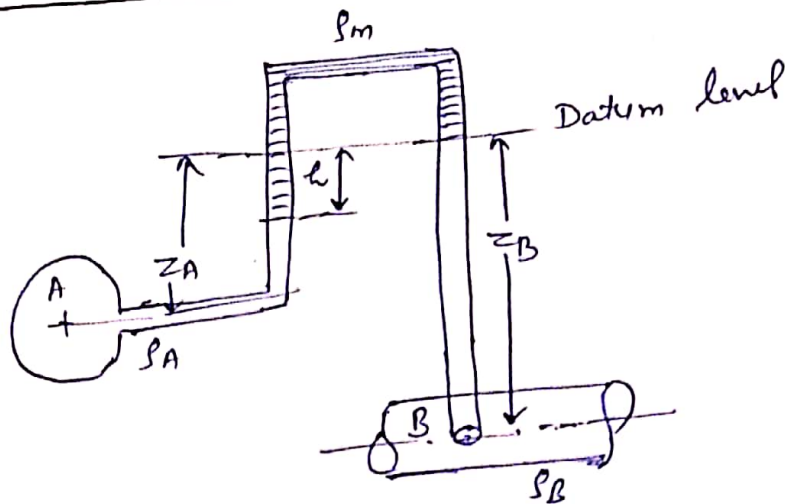
$$P_A - P_B = S_m g h + S g [z_B - z_A - h]$$

If datum levels are same then $\therefore P_A = P_B$

$$P_A - P_B = (S_m - S) g h$$

- In upright U-tube manometer lower part of manometer contains a manometric liquid which is heavier than liquid for which pressure difference is to be measured.

⑥ Inverted U-tube manometer



$$P_A - \rho g (z_A - h) - \rho_m g h + \rho g z_B = P_B$$

$$P_A - P_B = \rho_m g h + \rho g (z_A - h) - \rho g z_B$$

If same fluid and same height level

$$P_A - P_B = (\rho_m - \rho) g h$$

- In Inverted U-tube manometer, the manometric fluid is lighter than the working fluid. Hence difference in height is enhanced as compared to heavier fluid.
- Therefore, Inverted U-tube manometers are employed for accurate measurement or for small pressure difference between two point in liquids.

Notes:

- A micromanometer is used for very small pressure difference or for small pressure difference with high precision.
- mechanical gauges are used for measuring high pressure where high precision is not required. Here elastic element deflects under the action of applied pressure and movement is mechanically magnified.
- eg, Bourdon tube pressure gauge have elliptical bronze tube that becomes circular with application of internal pressure.

Tutorial problem (solved)

Calculate the gage pressure at A, B, C and D as shown in figure in N/m^2 , kgf/cm^2 and mWG .

Solution

Point A

$$P_A + 1000 \times 9.81 \times 0.6 = P_{\text{atm}}$$

$$P_A - P_{\text{atm}} = P_{\text{gage}}$$

$$= -1000 \times 9.81 \times 0.6$$

$$= -5886 \text{ N/m}^2$$

$$= -1.06 \text{ kgf/cm}^2$$

$$= \underline{\underline{-0.6 \text{ mWG}}}$$

Point B

$$P_B - 1000 \times 9.81 \times 0.5 = P_{\text{atm}}$$

$$(P_{\text{gage}})_B = 4905 \text{ N/m}^2$$

$$= 0.05 \text{ kgf/cm}^2$$

$$= \underline{\underline{0.5 \text{ mWG}}}$$

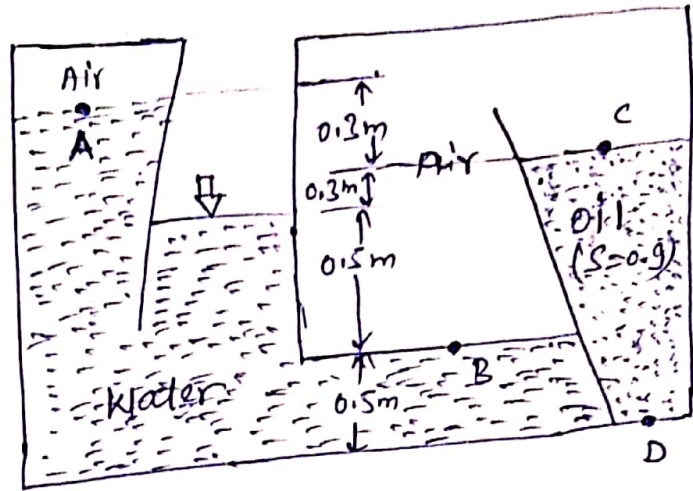
Point C

$$P_C + 0.8 \times \left(\begin{array}{l} P_{\text{air}} \times 9.81 \\ \rightarrow \text{neglected} \end{array} \right) + 0.5 \times 1000 \times 9.81 - 1.0 \times 1000 \times 9.81 = P_{\text{atm}}$$

$$(P_{\text{gage}})_C = 4905 \text{ N/m}^2$$

$$= 0.05 \text{ kgf/cm}^2$$

$$= \underline{\underline{0.5 \text{ mWG}}}$$



Point D

$$(P_D)_{\text{atm}} - 0.9 \times 1000 \times 9.81 \times 1.3 = (P_C)_{\text{atm}}$$

$$(P_D)_{\text{atm}} = 4905 + 11477.7$$

$$= 16382.7 \text{ N/m}^2$$

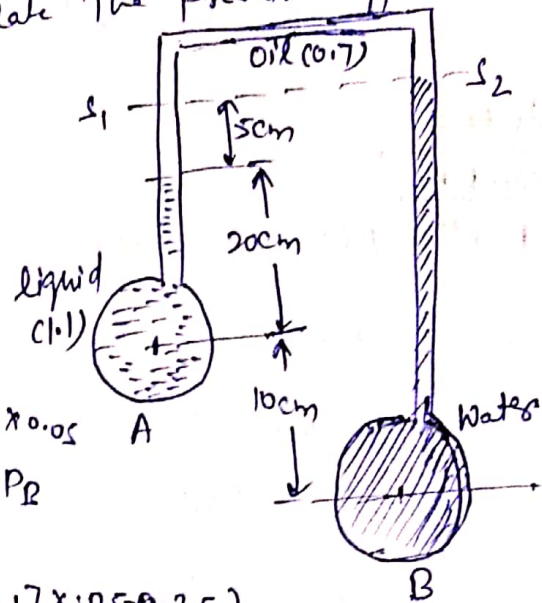
$$= 1.67 \text{ kgf/cm}^2$$

$$= \underline{\underline{1.67 \text{ mWG}}}$$

Tutorial problem (Solved)

An inverted U-tube is connected across two points A and B carrying fluids as shown. Calculate the pressure difference between centre lines of pipes.

Solution



$$P_A - 1.1 \times 1000 \times 9.81 \times 2 - 0.7 \times 1000 \times 9.81 \times 0.05 + 1000 \times 9.81 \times 0.35 = P_B$$

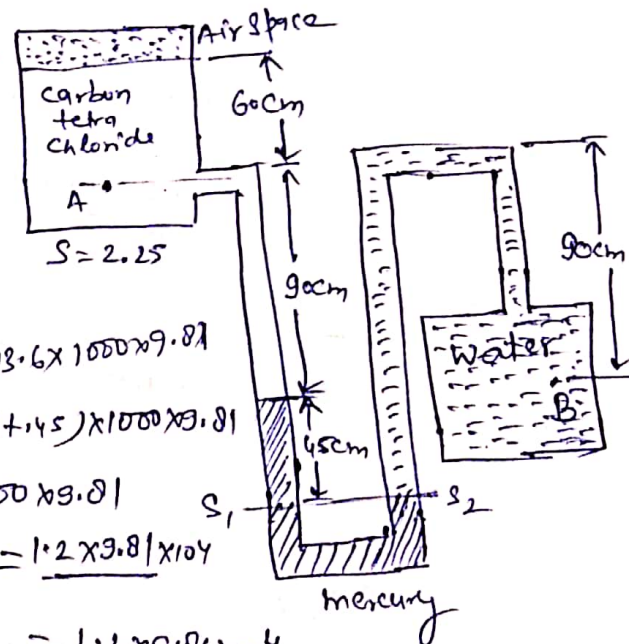
$$P_A - P_B = 1000 \times 9.81 (1.1 \times 2 + 0.7 \times 0.05 - 0.35)$$

$$= 1000 \times 9.81 (2.2 + 0.035 - 0.35)$$

$$= -931.95 \approx -932 \text{ N/m}^2$$

Tutorial problem (Solved)

A closed tank is filled with Carbon tetrachloride ($S=2.25$) and has a small air space at its top as shown in figure. A manometer is connected between point A and water tank as shown. If the pressure at point B in water is 1.2 kgf/cm^2 , find the pressure in KN/m^2 in the closed air space at the top of Carbon tetrachloride tank.



Solution

$$P_{\text{air}} + (0.6 + 0.9) \times 2.25 \times 1000 \times 9.81 - (0.9 + 0.45) \times 1000 \times 9.81 + 0.9 \times 1000 \times 9.81 = 1.2 \times 9.81 \times 10^4$$

$$P_{\text{air}} + 1000 \times 9.81 (1.5 \times 2.25 - 1.45) = 1.2 \times 9.81 \times 10^4$$

$$P_{\text{air}} = 9.81 \times 10^3 [12 - 9.045] = 28988 \text{ N/m}^2$$

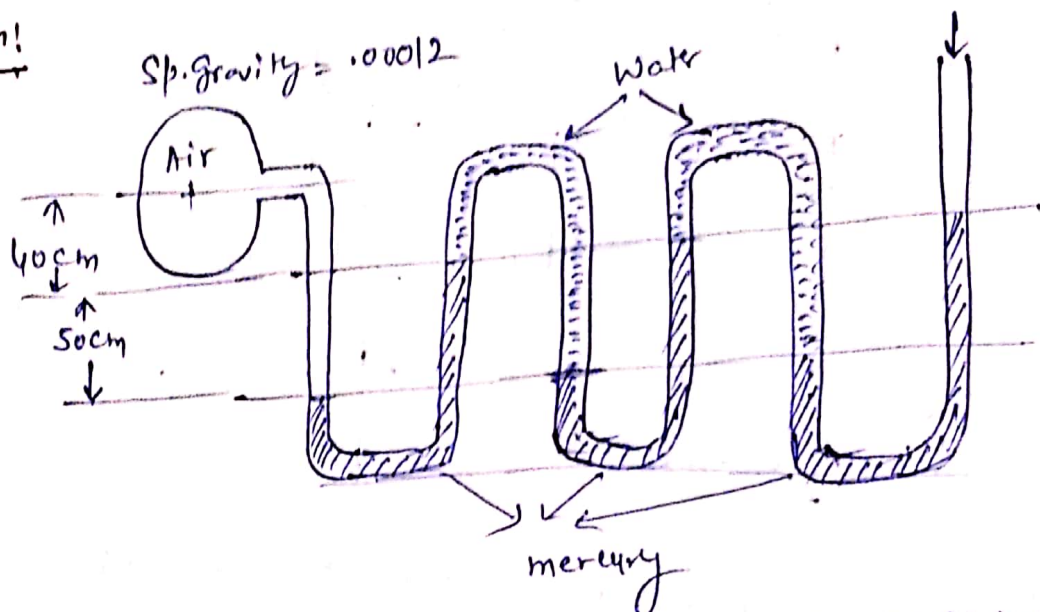
$$= 28.988 \text{ KN/m}^2$$

$$\approx 29 \text{ KN/m}^2$$

Tutorial problem (solved)

A multi-tube manometer is employed to determine pressure in a pipe. For the levels in manometer as shown, compute the pressure in the pipe. What would be the minimum length of a single mercury filled V-tube to record this pressure?

Solution!



$$P_{\text{Air}} + 0.0012 \times 10^3 \times 9.81 \times 0.9 - \frac{0.5 \times 1000 \times 12.6 \times 9.81}{\times 3} + \frac{0.5 \times 10^3 \times 9.81}{\times 2} = 0$$

$$\begin{aligned} P_{\text{Air}} &= 9.81 \times 10^3 \left[0.5 \times 12.6 \times 3 - 0.5 \times 2 - 0.0012 \times 0.9 \right] \\ &= 9.81 \times 10^3 \left[20.4 - 1 - 0.00108 \right] \\ &= 190.31 \text{ kN/m}^2 \end{aligned}$$

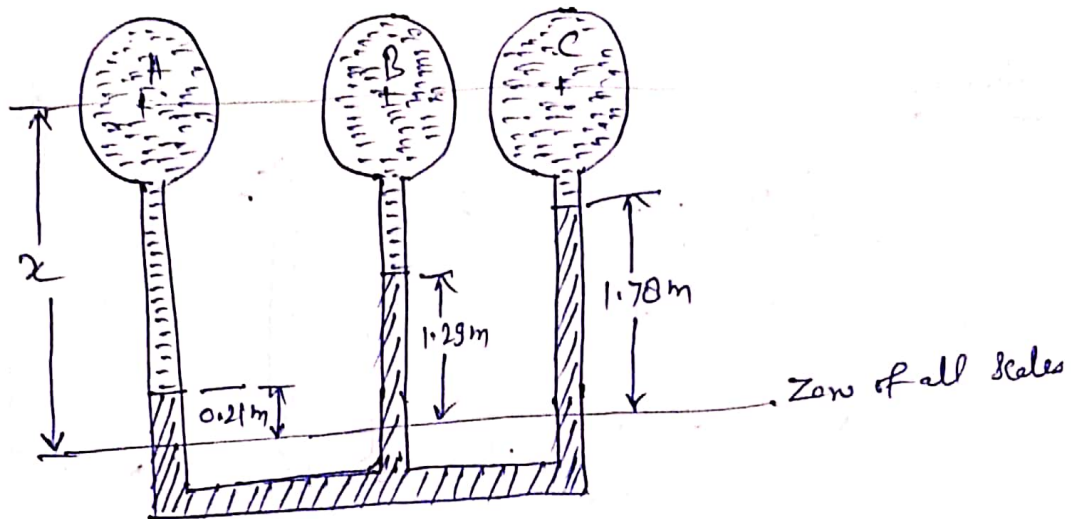
Now length of single V-tube with mercury filled

$$\begin{aligned} &= \frac{190.31 \times 10^3}{12.6 \times 1000 \times 9.81} \\ &= \underline{\underline{142.64 \text{ cm}}} \end{aligned}$$

Tutorial problem (solved)

For multiple differential manometers point A, B and C are at the same elevation. Find $P_A - P_B$, $P_A - P_C$, $P_B - P_C$ in terms of water column.

Solution



$$\frac{P_A}{\rho g} + (x - 0.21) + 13.6 \times 0.21 - (1.29 - 0.21) \times 13.6 - (x - 1.29) = \frac{P_B}{\rho g}$$

$$\frac{P_A - P_B}{\rho g} = - (1.29 - 0.21) + 13.6 \times (1.29 - 0.21) = 1.08 \times 12.6 = \underline{13.608 \text{ m water}}$$

$$\frac{P_A}{\rho g} + (x - 0.21) + 13.6 \times 0.21 - 1.78 \times 13.6 - (x - 1.78) = \frac{P_C}{\rho g}$$

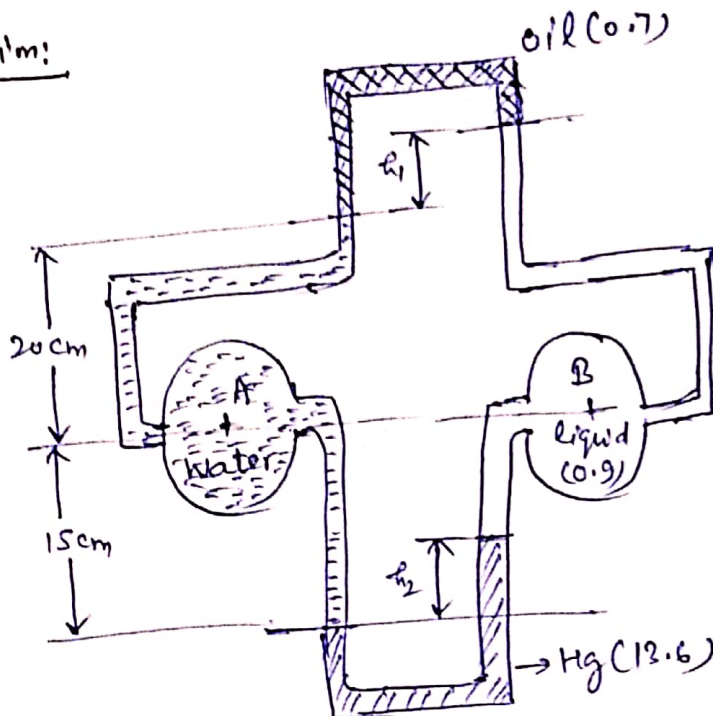
$$\frac{P_A - P_C}{\rho g} = - (1.78 - 0.21) + 13.6 \times (1.78 - 0.21) = 1.57 \times 12.6 = \underline{19.782 \text{ m water}}$$

$$\frac{P_B - P_C}{\rho g} = \frac{P_A - P_C}{\rho g} - \frac{P_A - P_B}{\rho g} = 19.782 - 13.608 = \underline{6.174 \text{ m of water}}$$

Tutorial problem (solved)

Two V-tube manometers one upright and other inverted type are connected across a water line and an oil line as shown in figure. If $h_1 = 5\text{cm}$, what shall h_2 be?

Solution:



Upright V-tube

$$\frac{P_A}{\rho g} + 15 - 13.6 \times h_2 - (15 - h_2) \times 0.9 = \frac{P_B}{\rho g}$$

$$\begin{aligned} \frac{P_A - P_B}{\rho g} &= (13.6 - 0.9) h_2 + 0.9 \times 15 - 15 \\ &= 12.7 h_2 - 10.15 \end{aligned}$$

Inverted V-tube

$$\frac{P_A}{\rho g} - 0.2 - h_1 \times 0.7 + (h_1 + 0.2) \times 0.9 = \frac{P_B}{\rho g}$$

$$\begin{aligned} \frac{P_A - P_B}{\rho g} &= 0.7 h_1 - 0.9 h_1 + 0.2 - 0.2 \times 0.9 \\ &= -0.01 + 0.02 = 0.01 \end{aligned}$$

$$(h_1 = 0.05\text{m})$$

$$\Rightarrow 12.7 h_2 - 10.15 = 0.01$$

$$h_2 = 0.19685\text{cm}$$

$$\approx \underline{\underline{0.197\text{cm}}}$$

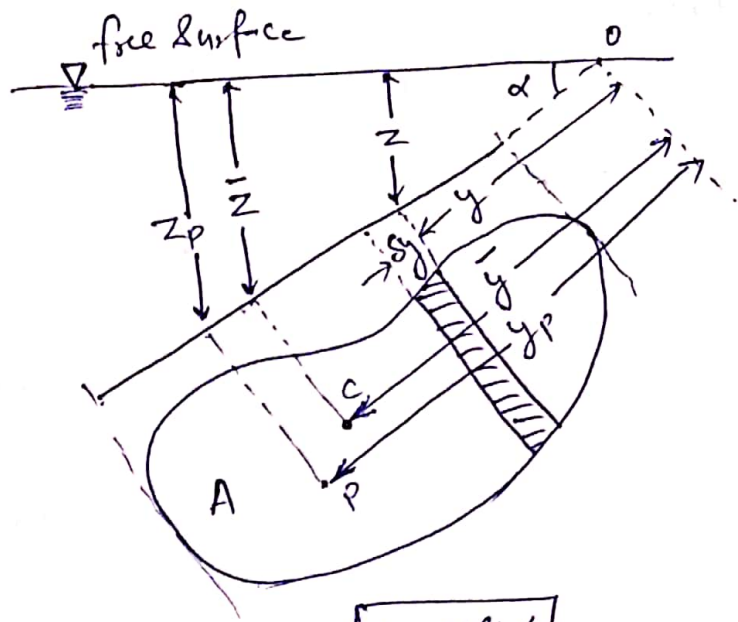
Hydrostatic force on submerged plane surface

Let a plane surface of area A with its centroid at C be placed at an angle α with respect to free surface of liquid.

Consider elementary shaded strip at a depth z .

$$\text{Pressure } p \text{ on the strip} = \rho g z$$

$$\begin{aligned}\text{Force on the strip} \\ &= \delta F = p \delta A \\ &= \rho g z \delta A \\ &= \rho g y \sin \alpha \delta A\end{aligned}$$



Force on entire surface or total force

$$z = y \sin \alpha$$

$$= F = \int dF = \int p dA$$

$$\begin{aligned}&= \int \rho g y \sin \alpha dA = \rho g \sin \alpha \int y dA \\ &= \rho g \sin \alpha \bar{y} A\end{aligned}$$

$$\bar{y} = \frac{\int y dA}{A}$$

$$F = \rho g \bar{z} A$$

↳ Total force or hydrostatic force on a plane surface.

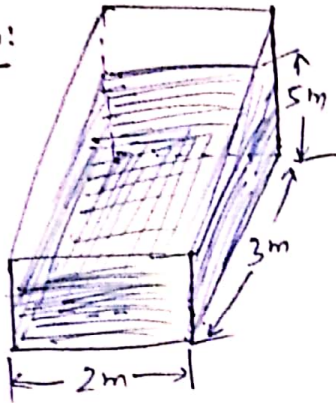
It equals pressure at centroid of the surface times the submerged area A of the surface.

This acts normal to the surface.

Tutorial problem (Solved)

A box of rectangular base $2\text{m} \times 3\text{m}$ contains gasoline (Sp. gravity 0.7) upto a height of 5m . Calculate the force on the base and on each vertical face. Locate their lines of action.

Solution:



Base

$$\begin{aligned}\text{Force} &= \rho g \bar{z} A \\ &= 1000 \times 0.7 \times 9.81 \times 5 \times 2 \times 3 \\ &= 206010 \text{ N} = 206 \text{ kN}\end{aligned}$$

Pressure is uniform over base

$$z_p = \bar{z}$$

i.e. resultant must pass through centroid of base and directed vertically downwards.

Each face $2\text{m} \times 5\text{m}$

$$\bar{z} = \frac{5}{2} = 2.5 \text{ m}$$

$$\text{Force} = F = \rho g \bar{z} A$$

$$\begin{aligned}&= 1000 \times 0.7 \times 9.81 \times 2.5 \times 2 \times 5 \\ &= 171675 \text{ N} \approx 172 \text{ kN}\end{aligned}$$

Each face $3\text{m} \times 5\text{m}$

$$z = \frac{5}{2} = 2.5 \text{ m}$$

$$\text{Force} = F = \rho g \bar{z} A$$

$$\begin{aligned}&= 0.7 \times 1000 \times 9.81 \times 2.5 \times 3 \times 5 \\ &= 257513 \text{ N} \approx 258 \text{ kN}\end{aligned}$$

Centre of pressure for each vertical face is at depth $\frac{2}{3}$ times the submerged height

$$= \frac{2}{3} \times 5 = 3.33 \text{ m below the free surface level.}$$

Buoyancy

If a body is submerged in a fluid, wholly or partially, the body experiences an upward force due to the fluid surrounding it.

" The phenomenon of force exerted by fluid on the body is called buoyancy and the force is called 'buoyant force'."

Buoyancy force experienced by body can be estimated by Archimedes principle of buoyancy.

Archimedes principle of buoyancy

Statement " A body immersed in a fluid (partially or wholly) experiences an upward buoyancy force equivalent to the weight of the fluid displaced by the body."

⇒ A body floats partially immersed in a fluid to the extent that the weight of the fluid displaced equals the weight of the body.

F_B = upward buoyancy force

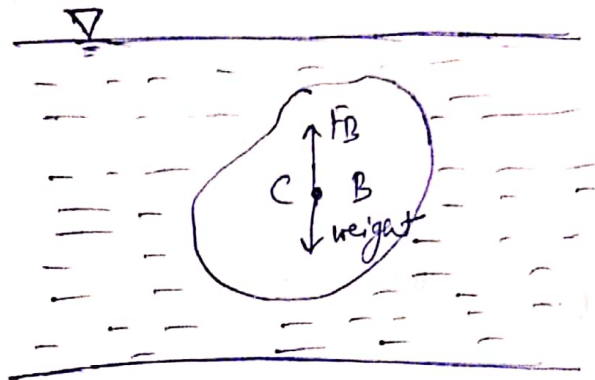
$$F_B = \rho g V$$

(weight of the volume V of the fluid displaced)

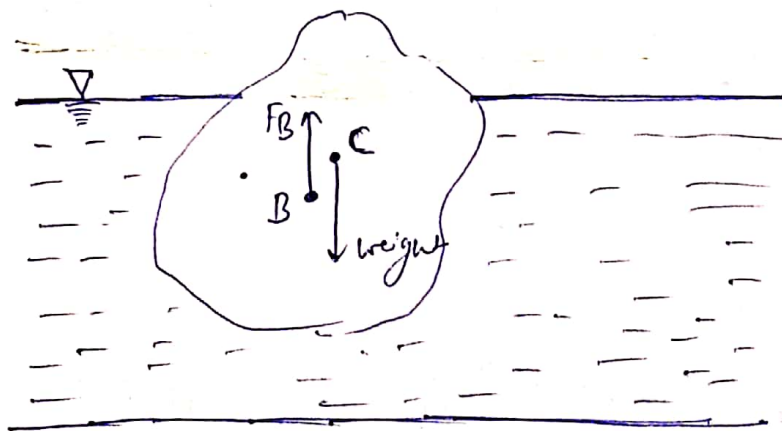
Centre of Buoyancy (point B)

The point of application of buoyant force is known as centre of buoyancy.

- ① In case of fully submerged body, centre of buoyancy and centroid of body coincide.

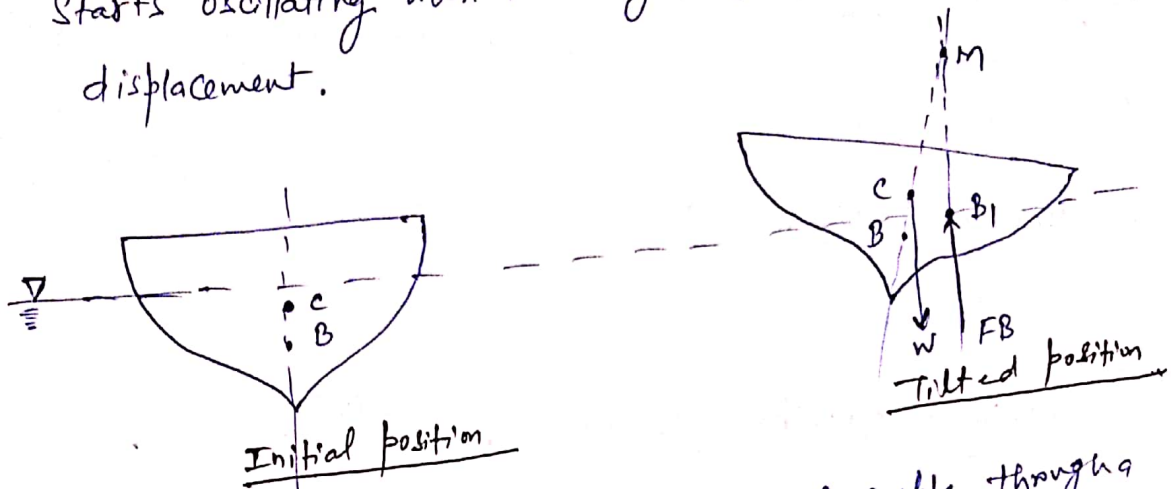


- ② In case of partially submerged body, centre of buoyancy and centroid of body does not coincide, centre of buoyancy is at the centre of displaced fluid.



Metacentre (Point M) and metacentric height of a floating body

- When a floating body is given a tilt, it is set in a state of oscillation as if suspended at the metacentre.
- Hence metacentre is the point about which a floating body starts oscillating when the body is given small angular displacement.



When a floating body is tilted by an external couple, through a small angle, the centre of buoyancy will change to B_1 .

"The point of intersection (M) of original line through B and C (produced) and vertical through new centre of buoyancy (B_1) is called metacentre."

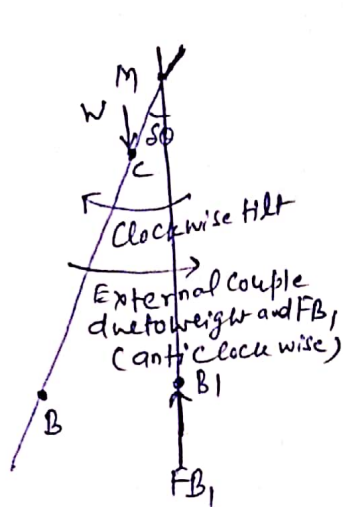
The distance CM, measured from centre of gravity, C of the body to metacentre, M is called metacentric height.

Equilibrium of Submerged and floating bodies

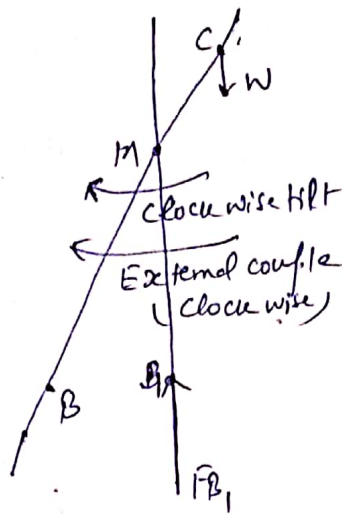
- General conclusion from study of mechanics says that position of equilibrium of a body is the position of minimum potential energy corresponding to equilibrium state. Therefore, centre of gravity, C of the body should be as low as possible.

Conditions of equilibrium of floating bodies

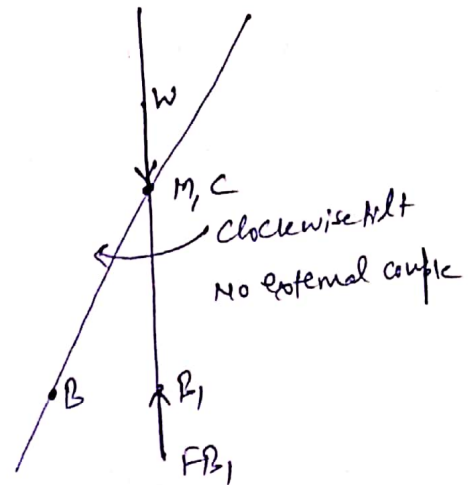
metacentric height determines the conditions of equilibrium for floating bodies.



Stable equilibrium.
M above C
+ve metacentric height



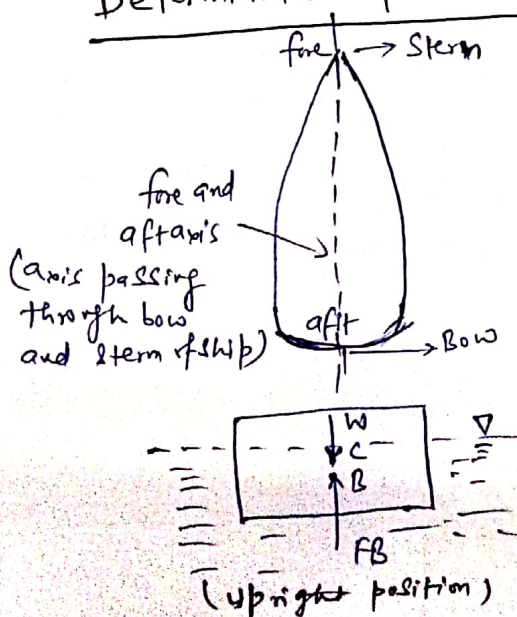
Unstable equilibrium.
M below C
-ve metacentric height



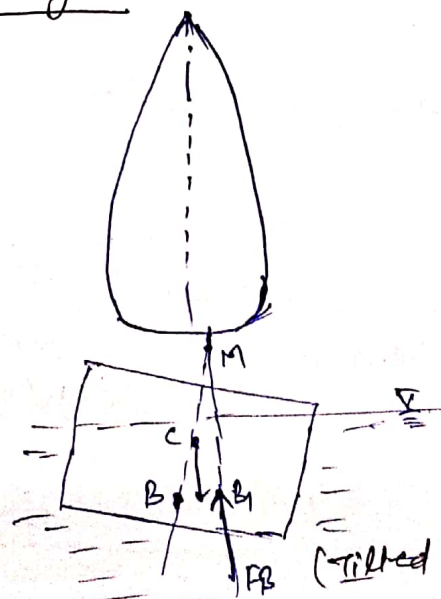
Neutral equilibrium
M coincides C
Zero metacentric height

- For stability of a floating body, its metacentric height should be positive. Larger the metacentric positive height, more stable is the body.

Determination of metacentric height



Consider a floating body with



plan form symmetrical about fore and aft axis.
(top view)

Theoretically

$$B.M = \frac{I}{V}$$

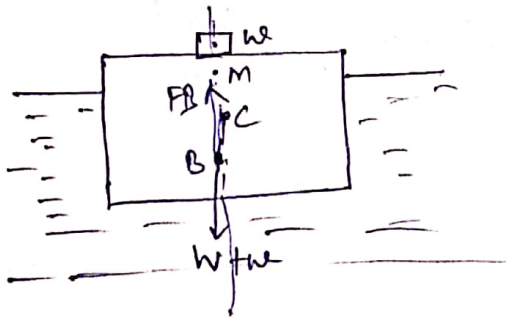
I = Area moment of inertia of the top view area at the liquid level about fore and aft axis i.e. axis of tilt

V = volume of the liquid displaced by the body.

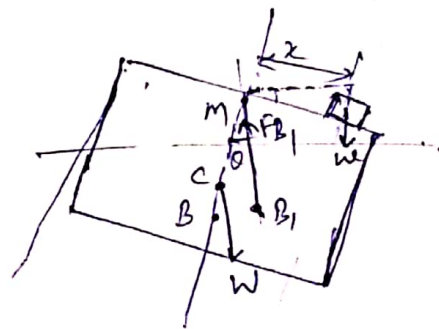
$$\text{metacentric height} = CM = BM - BC = \frac{I}{V} - BC$$

$$CM = \frac{I}{V} - BC$$

Experimentally



(Initial position)



(Tilted position)

Let W = weight of ship.

w = experimental movable weight (initially at the centre of deck)

Buoyancy force F_B balances the weight

$$F_B = w + W$$

When the weight (w) is moved through a distance x which causes a tilt θ . In tilted position, body is in equilibrium under following forces and couples

$$w + W = F_B$$

taking moment about m

$$w \times x \cos \theta = W \times CM \sin \theta$$

$$CM = \frac{w}{W} \frac{x}{\tan \theta}$$

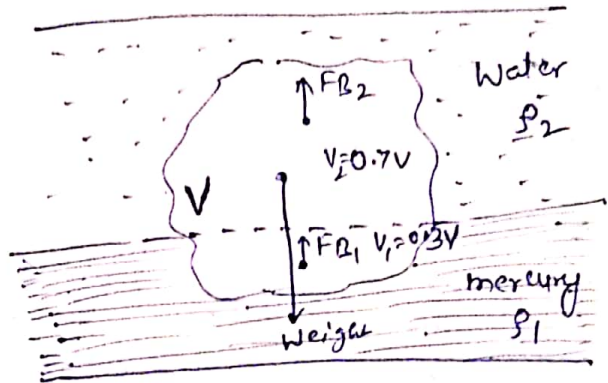
Tutorial problem (Solved)

A metallic body floats at the interface of mercury (Sp. grav. 12.6) and water such that 30% of its volume is submerged in mercury and 70% in water. Estimate the density of metal.

Solution

Buoyancy force due to mercury

$$\begin{aligned} F_{B1} &= \rho g V \\ &= 12.6 \times 1000 \times 9.81 \times 0.3V \\ F_{B1} &= 4080 g V \end{aligned}$$



$$\begin{aligned} \text{Buoyancy force due to water} &= F_{B2} = \rho g V \\ &= 1000 \times 9.81 \times 0.7V \\ &= 700 g V \end{aligned}$$

$$\begin{aligned} \text{Total buoyancy force} &= F_B = F_{B1} + F_{B2} \\ &= (4080 + 700) g V \\ &= 4780 g V \end{aligned}$$

Total buoyancy force must be equal to weight of body

$$\text{Wt. of body} = \rho g V$$

equating

$$\rho g V = 4780 g V$$

$$\underline{\underline{\rho = 4780 \text{ kg/m}^3}} \quad (\text{density of metallic body})$$

Tutorial Problem (Solved)

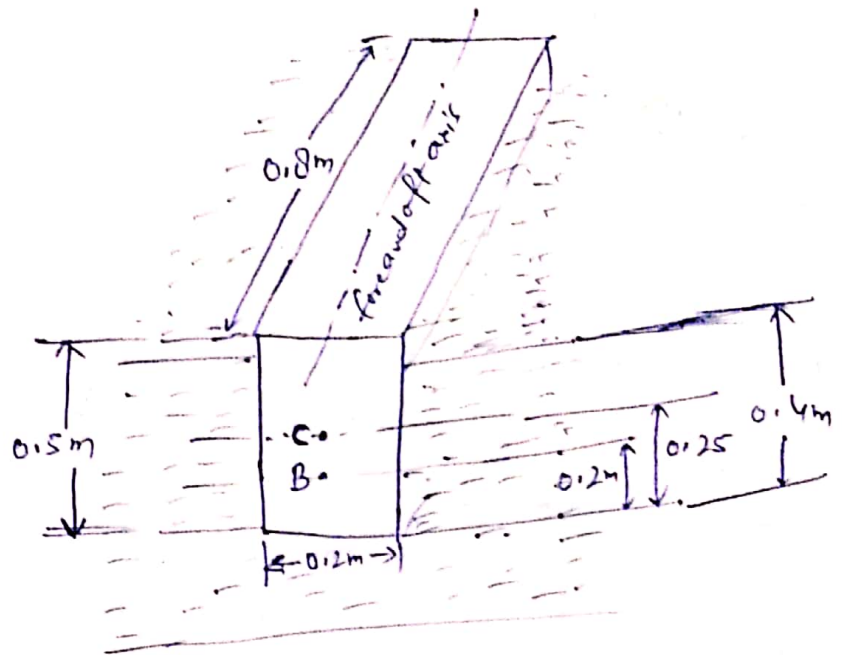
A block of wood $0.2\text{m} \times 0.5\text{m}$ in cross section and 0.8m long has a mass of 64kg . Can the block float with 0.5m side vertical in water? Check the stability with 0.2m side vertical.

Solution

$$\begin{aligned}\text{Volume of wood block} \\ &= 0.2 \times 0.5 \times 0.8 \\ &= 0.08\text{ m}^3\end{aligned}$$

$$\begin{aligned}\text{density of wood} \\ \text{block} &= \frac{64}{0.08} = 800\text{ kg/m}^3 \\ &< 1000\text{ kg/m}^3 \text{ (density of water)}\end{aligned}$$

hence block will float in water.



Now weight of volume of water displaced must be equal to weight of block.

$$\begin{aligned}1000 \times g \times V &= 64g \\ V &= \frac{64}{1000} = 0.064\text{ m}^3 \text{ (Volume of water displaced)}\end{aligned}$$

With possible floating with 0.5m height vertical, the height of submergence is $= \frac{0.064}{0.8 \times 0.2} = 0.4\text{m}$

Centre of buoyancy will be at the centroid of displaced water i.e. at 0.2m from bottom of block.

$$BC = 0.25 - 0.02 = 0.05\text{m}$$

$$CM = \frac{I}{V} - BC$$

I = Area moment of inertia of top view area about fore and aft axis.

$$= \frac{0.8 \times (0.2)^3}{12} = 5.33 \times 10^{-4}$$

$$CM =$$

$$\begin{aligned}\frac{5.33 \times 10^{-4}}{0.064} - 0.05 &= -0.04167\text{m} \\ CM &< 0 \text{ (negative)} \\ \text{(Hence unstable floating for } 0.5\text{m side vertical)}\end{aligned}$$

Checking stability with 0.2 m side vertical

with 0.2 m height vertical, the height of submergence is

$$= \frac{0.064}{1.8 \times 1.5}$$

$$= 0.023$$

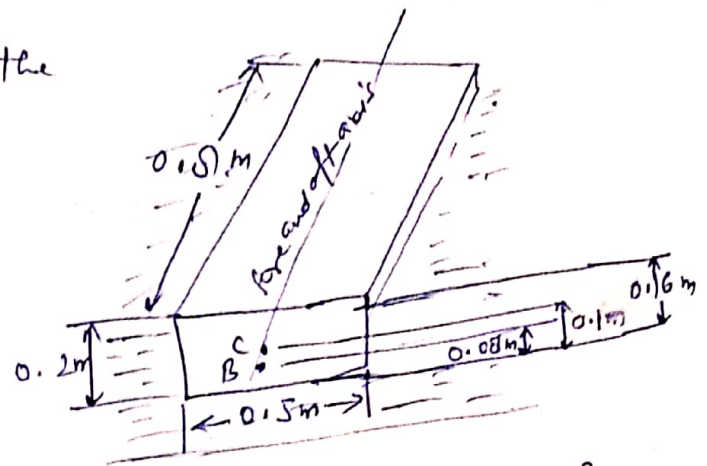
⇒ B point is at 0.08 m from base

$$C_m = \frac{I}{V} - BC$$

$$I = \frac{1.8 \times 1.5^3}{12} = 8.22 \times 10^{-2}$$

$$C_m = \frac{8.22 \times 10^{-2}}{0.064} - 0.02 = 1.1 \text{ m (true)}$$

Hence stable floatation for 0.2 m side vertical.



$$BC = 0.1 - 0.08 = 0.02 \text{ m}$$

Tutorial problem (solved)

A solid cylinder of diameter 30 cm and height 15 cm floats with its axis vertical in sea water (sp. gravity 1.03). If the specific gravity of cylinder material is 0.9. Examine the stability of floatation with its axis vertical.

Solution

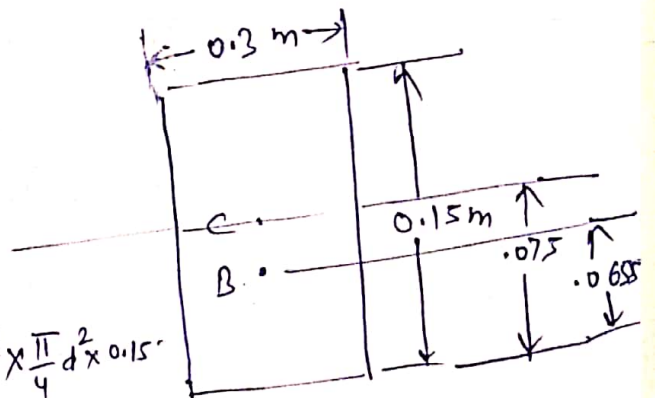
Weight of volume of fluid displaced

= weight of cylinder

$$1.03 \times 1000 \times g \times V = 0.9 \times 1000 \times g \times \frac{\pi}{4} d^2 \times 0.15$$

$$V = 0.032767 \pi d^2 \text{ m}^3$$

$$= 9.2647 \times 10^{-3} \text{ m}^3$$



$$\text{Height of Submergence} = \frac{0.032767 \pi d^2}{\pi/4 \times (d)^2}$$

$$= 0.1311 \text{ m}$$

Hence B point is = 0.0655 m from base

$$Bc = 9.5 \times 10^{-2} \text{ m}$$

$$I = \frac{\pi}{64} d^4 = \frac{\pi \times (1.3)^4}{64} = 3.976 \times 10^{-4}$$

$$C_m = \frac{I}{V} - Bc$$

$$= \frac{3.976 \times 10^{-4}}{9.2647 \times 10^{-2}} - 9.5 \times 10^{-2}$$

$$= 0.0334 \text{ m} = 3.34 \text{ cm}$$

Hence stable equilibrium of floatation.

Tutorial problem (Solved)

A ship of 3000 tonne has a weight of 75 tonne moved 6 m across the deck and it inclines the ship $4\frac{1}{2}^\circ$. Find metacentric height.

Solution

$$C_m = \frac{w_e}{W} \times \frac{2}{\tan \theta}$$

$$= \frac{75}{3000} \times \frac{6}{\tan 4\frac{1}{2}^\circ}$$

$$= 1.906 \text{ m}$$

Fluid Kinematics

"Fluid Kinematics deals with study of velocity and acceleration of fluid particles and their distribution in space without consideration of any forces involved."

velocity and acceleration at a point in a fluid can be denoted as

$$\text{Velocity vector} = \vec{V} = u\vec{i} + v\vec{j} + w\vec{k}$$

where u , v and w are the components of velocity in x , y and z directions, respectively.

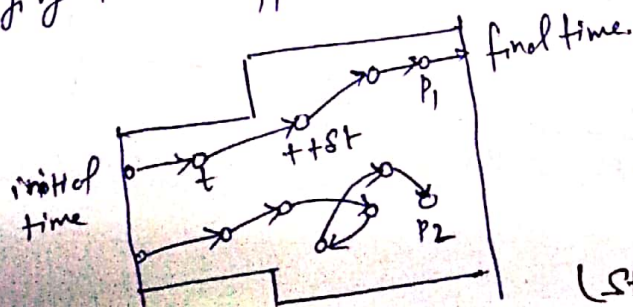
$$\text{Similarly } \vec{a} = a_x\vec{i} + a_y\vec{j} + a_z\vec{k}$$

where a_x , a_y and a_z are the rectangular components of acceleration in x , y and z directions, respectively.

Lagrangian and Eulerian approach

Fluid motion can be studied using two approaches.

In Lagrangian approach, fluid particle is identified and followed during course of motion with time. The particle may change its shape, size and state as it moves. Difficulty in tracing a single fluid particle rules out the possibility of applying this approach.



(Study of each particle with time)
(Lagrangian approach)

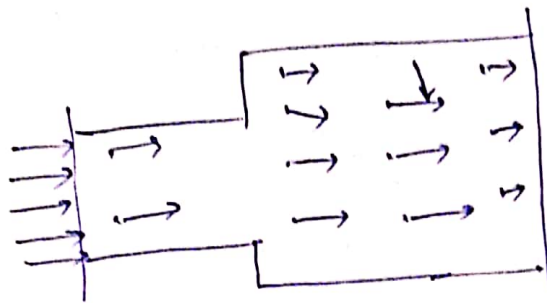
In Eulerian approach fluid motion is studied at fixed points in flow field. motion of fluid is specified by velocity components as a function of space and time.

$$u = u(x, y, z, t)$$

$$v = v(x, y, z, t)$$

$$w = w(x, y, z, t)$$

This approach is applied in study of fluid mechanics



Study at all points
a time
(Eulerian approach)

Types of fluid flow

Steady and Unsteady flow

A flow is considered to be steady if dependent fluid variables (eg. velocity, density, viscosity, pressure and temperature) at a point in flow do not change with time.

i.e.

$$\frac{\partial (\text{dependent fluid variable})}{\partial t} = 0$$

$$\Rightarrow \frac{\partial u}{\partial t} = \frac{\partial v}{\partial t} = \frac{\partial w}{\partial t} = 0 = \frac{\partial p}{\partial t} = \frac{\partial \rho}{\partial t}$$

A flow is considered as unsteady if the dependent fluid variables at any point in flow changes with the passage of time.

$$\frac{\partial (\text{dependent fluid variable})}{\partial t} \neq 0$$

- Steadiness of flow implies that flow pattern does not change with time.
- Unsteadiness refers to changing flow pattern with passage of time at same points in space.

Uniform and Non uniform flow

A flow is said to be uniform at any instant of time if the velocity components are identical throughout the flow field. i.e. velocity components are same at different locations in flow.

$$\text{i.e. } \frac{\partial u}{\partial x} = \frac{\partial u}{\partial y} = \frac{\partial u}{\partial z} = \frac{\partial v}{\partial x} = \frac{\partial v}{\partial y} = \frac{\partial v}{\partial z} = 0$$

The flow is considered non uniform if velocity components at given instant of time are different at different locations.

$$\frac{\partial (\vec{V})}{\partial s} \neq 0$$

- "Conventionally uniformity of flow is defined in terms of velocity components rather than all dependent fluid variables"

- Steadiness refers to constancy with respect to "time" and Uniformity refers to constancy with respect to "space"

- Flow can be steady or unsteady independently of being its uniform or non-uniform.

- All four combinations are possible.

(a) Steady and uniform: Flow in straight long pipe of constant diameter with constant discharge.

(b) Steady and non-uniform: flow in a conical pipe having constant discharge.

(c) Unsteady and uniform: flow in a straight long pipe of constant diameter with varying discharge.

(d) Unsteady and nonuniform: flow in a conical pipe having varying discharge.

One, two and three dimensional flow

"A fluid flow is said to be one, two and three dimensional if Velocity component is the function of one, two and three coordinates, respectively."

Symbolically

Steady

Unsteady

One-dimensional flow: $\bar{V} = f(x)$

$$\bar{V} = f(x, t)$$

Two dimensional flow: $\bar{V} = f(x, y)$

$$\bar{V} = f(x, y, t)$$

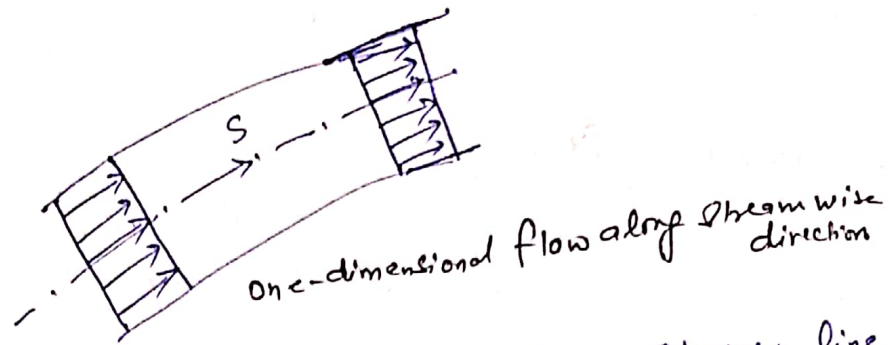
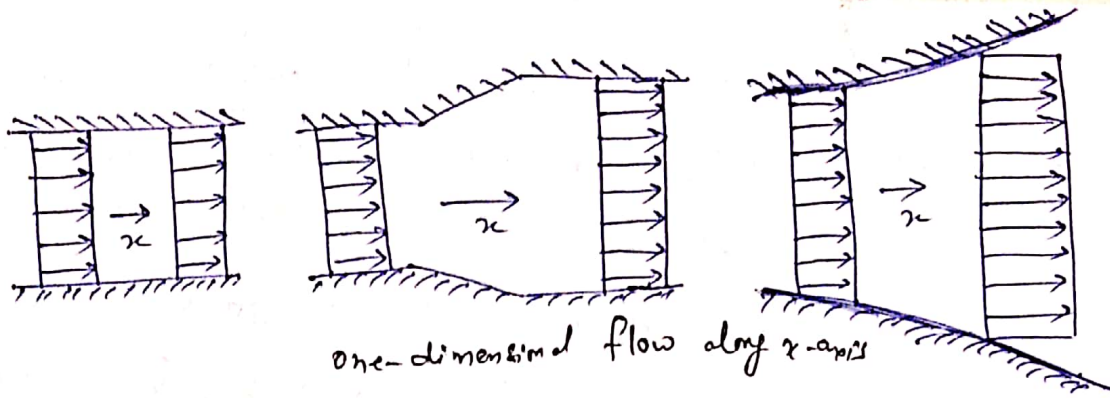
Three dimensional flow: $\bar{V} = f(x, y, z)$

$$\bar{V} = f(x, y, z, t)$$

One-dimensional flow

- In this flow the dependent variables are only function of one space coordinate eg. x .

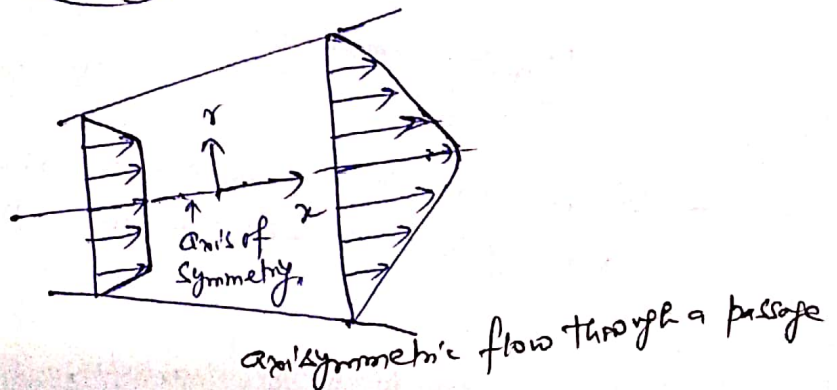
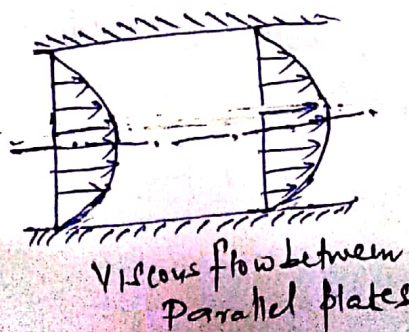
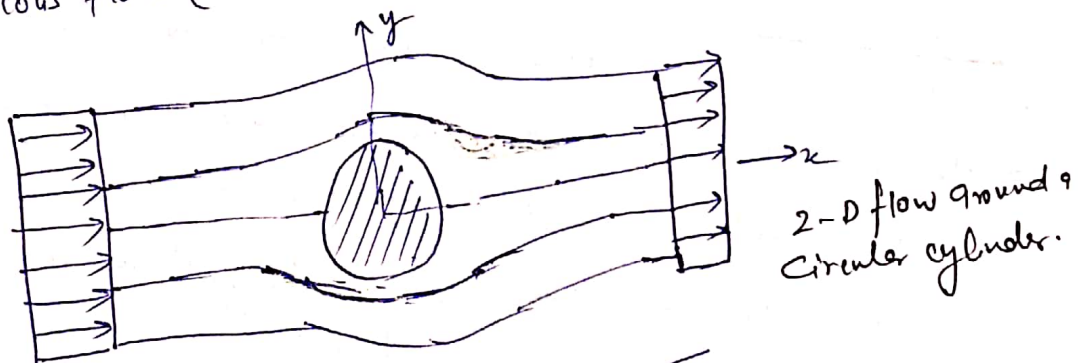
- eg. flow through pipes, channels and variable area ducts if velocity distribution is considered constant at each cross section.



- The axis of flow does not have to be a straight line for a flow to be one dimensional.

Two-dimensional flow

- In this flow dependent variables are function of two space coordinates.
- This means flow is identical in parallel planes.
- examples are flow around a circular cylinder of infinite length, viscous flow (real fluid) through pipes, between parallel plates etc.



- A flow is called axisymmetric if velocity profile is symmetrical about axis of symmetry. i.e. velocity profile is same at different diametral planes drawn through passage, i.e. No variation of velocity profile with respect to θ
- An axisymmetric flow is essentially 2-D flow, $\bar{V} = f(r, x)$

Three-dimensional flow

- In three dimensional flows flow properties vary in all three directions such as flow in a river, flow within fluid machines etc.

Laminar flow and turbulent flows

A Laminar flow is a flow in which fluid particles move in thin layers (laminae) with one layer sliding over other. The particles move in definite and observable paths. There is no exchange of fluid particles from one layer to another. This flow is characterized by small velocity,

A turbulent flow is a flow in which all fluid particles are distributed and there is continuous transfer of momentum to adjacent layers. In this flow, the velocity varies from point to point in magnitude and direction as well as from time to time.

Two flows are characterized by Reynold's number

$$\text{Reynolds number (Re)} = \frac{\text{Inertia force}}{\text{Viscous force}}$$

- Viscous force is frictional force due to viscosity of fluid and try to make fluid motion in parallel layers.
- Inertia force is due to mass and velocity of fluid particles and try to diffuse ^{the} particles.

- At low velocities, the inertia forces are less in comparison to viscous forces and the flow is laminar.
- But when the velocity is high, fluid layers break due to predominance of inertia forces and flow is turbulent.

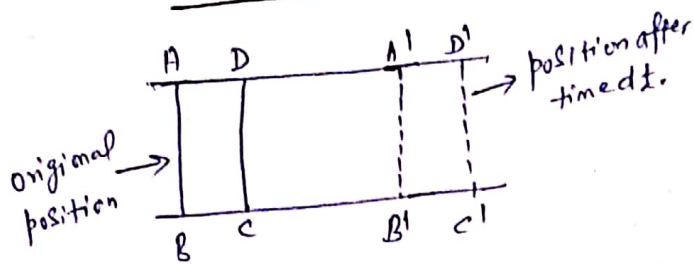
for pipe flow

$Re < 2100$ (laminar flow)

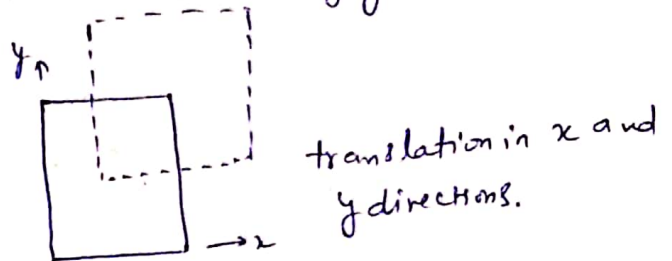
$Re > 3000$ (turbulent flow)

Different types of displacement of fluid particles

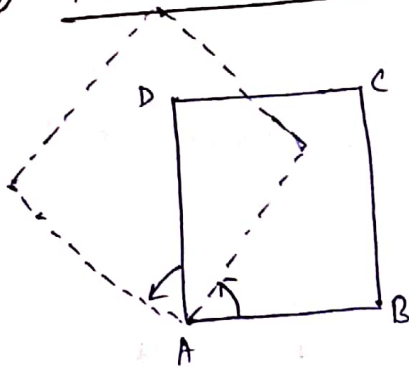
(a) Linear translation or pure translation or irrotational translation



- The element moves as a body without changing its dimensions.

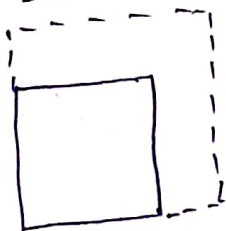


(b) Pure rotation or rotational translation

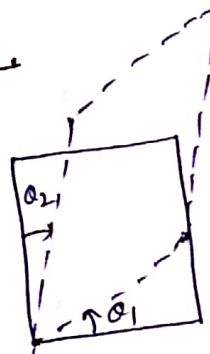


Rotation of AB and AD are in same direction.

(c) Pure distortion or deformation



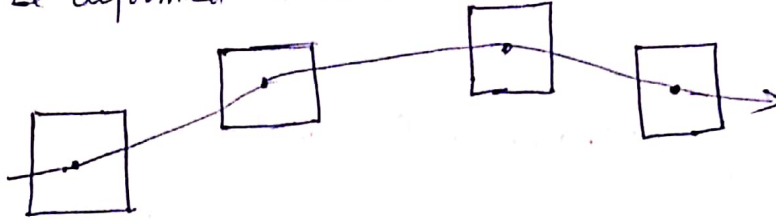
Volume or linear deformation



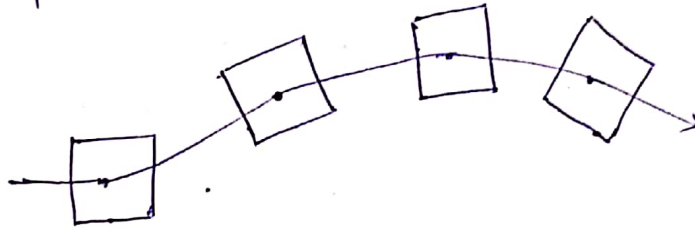
Angular or shear deformation

Rotational flow and Irrotational flow

An irrotational flow is one in which fluid element has no net rotation about its own axis which is perpendicular to plane of motion while moving in the direction of flow. The fluid element may be deformed or distorted.



If net rotation of fluid element about its own axis is not zero and they rotate while moving in the direction of flow, then the flow is termed as rotational flow.

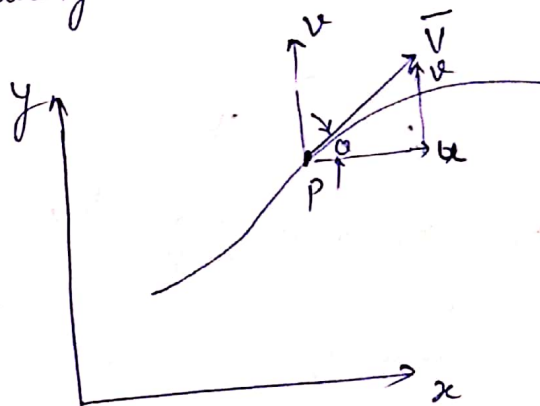


Streamlines, Pathlines and Streaklines

"A Streamline in a fluid flow is a line tangent to which at any point is the direction of velocity at that point at that instant."

By definition

velocity vector $\vec{V}$ at point P must be tangential to streamline at that point.



$$\tan \theta = \frac{dy}{dx} = \frac{v \cdot dt}{u \cdot dt}$$

Stream-line in x-y plane

$$\boxed{\frac{dx}{u} = \frac{dy}{v}}$$

General equation of Streamline in three-dimensional flow

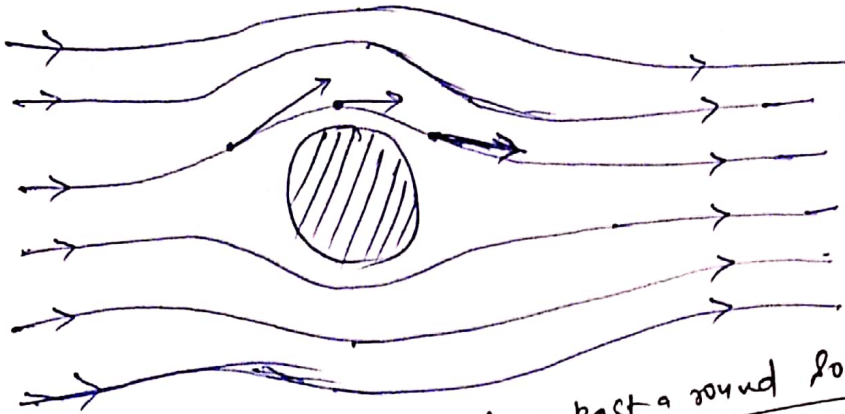
$dt = \text{small time interval}$

$$\boxed{\frac{dx}{u} = \frac{dy}{v} = \frac{dz}{w}}$$

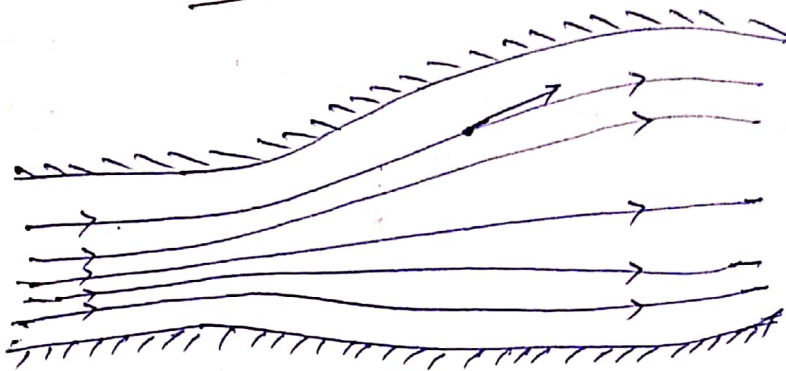
$$\frac{dx}{u} = \frac{dy}{v} = \frac{dz}{w}$$

Valid for

- Steady or unsteady
- Uniform or nonuniform
- viscous or inviscid
- compressible or incompressible



Streamlines of flow past a round solid object



Streamlines of flow within a close conduit

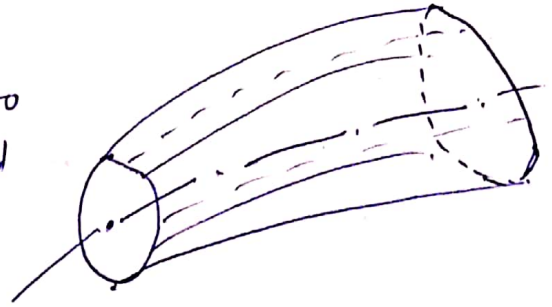
Some facts

- Streamlines are equivalent to instantaneous snap-shot indicating the direction of velocity in the entire flow field.
- Since the velocity is tangent at any point on the streamline, hence the component of velocity normal to streamline is zero, therefore there can be no flow across the streamline.
- No two streamlines can cross (intersect) each other because at the point of intersection fluid particle will have two velocity and that is physically impossible.

Stream tube

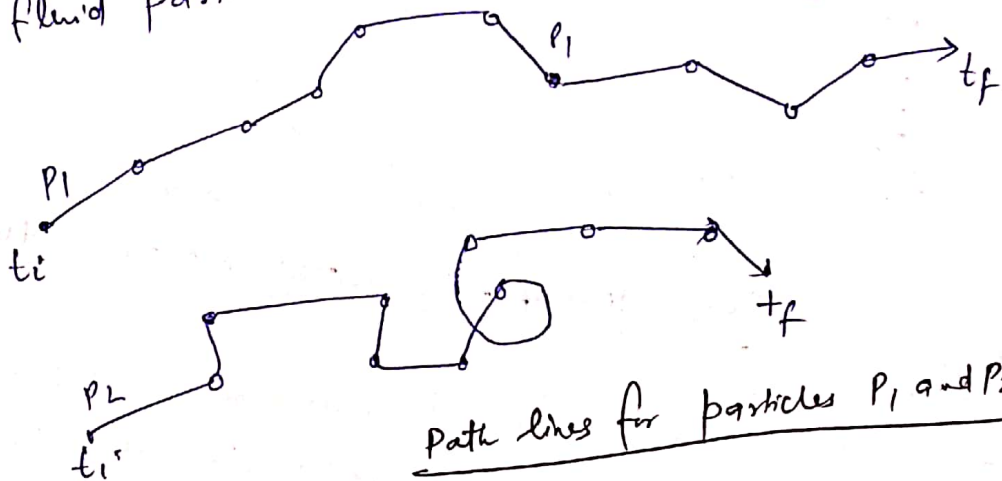
"The stream tube may be defined as a tubular space formed by collection of streamlines passing through the perimeter of a closed curve"

- Since the velocity component normal to streamlines is zero hence no fluid can enter or leave the stream tube except at ends.



Path line

"A path line in a fluid flow is the path traced by a single fluid particle in motion over a period of time." $(t_i \text{ to } t_f)$



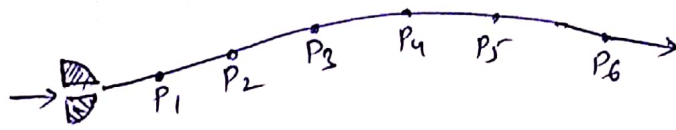
path lines for particles P_1 and P_2

- Tangent at a point on a path line must be the direction of velocity at that point at the time when the particle passes that point. A path line can therefore intersect itself at different times. In general path lines constitute a pattern of intersecting zig-zag lines.
- In steady flow velocity at each point remains unaltered with time. Hence streamlines pattern does not change with time. Individual fluid particles must follow the streamlines, hence for steady flow, streamlines and path lines coincide.

Streak line

"Streak line is the instantaneous picture of the positions of all the fluid particles that have passed through a fixed point in the flow field."

- example is lines formed by smoke particles ejected in the flow from a fixed nozzle is a streak line.



Streak line for particles
 $P_1, P_2, \dots$
(at $t = t_1$)

- In steady flow stream lines, path lines and streak lines coincide.

Tutorial problem (2d wed)

In a flow the velocity vector is given by.

$$\vec{V} = 3x\vec{i} + 4y\vec{j} - 7z\vec{k}$$

Determine the equation of stream line passing through a point (1, 4, 5).

Solution The equation of stream line is

$$\frac{dx}{u} = \frac{dy}{v} = \frac{dz}{w}$$

$$\frac{dx}{3x} = \frac{dy}{4y} = \frac{dz}{-7z}$$

Considering equation involving x and y

$$\frac{1}{3} \int \frac{dx}{x} = \frac{1}{4} \int \frac{dy}{y} \Rightarrow \frac{1}{3} \ln x = \frac{1}{4} \ln y + \ln C_1'$$

$C_1' = \text{a Constant of integration}$

$$\begin{aligned} \ln(x^{1/3}) &= \ln(y^{1/4}) + \ln C_1' \\ \ln x^{1/3} &= \ln(C_1' y^{1/4}) \\ &= \ln \frac{y^{1/4}}{C_1'} \Rightarrow \boxed{y = C_1 x^{4/3}} \end{aligned}$$

$C_1 = \text{another Constant}$

Considering equation involving x and z

$$\frac{dx}{3x} = \frac{dz}{-7z} \Rightarrow \frac{1}{3} \ln x = -\frac{1}{7} \ln z + \ln C_2'$$
$$\ln(x^{1/3}) = \ln(z^{-1/7}) + \ln C_2'$$

$$\ln(x^{1/3}) = \ln \frac{z^{-1/7}}{C_2'}$$

$$\boxed{z = C_2 x^{-7/3}}$$

$C_2 = \text{another Constant}$

point (1, 4, 5)

$$y = C_1 x^{4/3} \Rightarrow 4 = C_1 (1)^{4/3} \Rightarrow C_1 = 4$$

$$z = C_2 x^{-7/3} \Rightarrow 5 = C_2 (1)^{-7/3} \Rightarrow C_2 = 5$$

Hence stream line passing through (1, 4, 5) is

$$\boxed{\begin{aligned} y &= 4 x^{4/3} \\ z &= 5 x^{-7/3} \end{aligned}}$$

Velocity and Acceleration

Let $\vec{V} = u\vec{i} + v\vec{j} + w\vec{k}$ is the velocity vector at any point in the fluid flow.

Velocity components are the functions of space and time.

$$u = u(x, y, z, t)$$

$$v = v(x, y, z, t)$$

$$w = w(x, y, z, t)$$

Using differential calculus, infinitesimal change in u is given by:

$$\delta u = \frac{\partial u}{\partial x} \delta x + \frac{\partial u}{\partial y} \delta y + \frac{\partial u}{\partial z} \delta z + \frac{\partial u}{\partial t} \delta t$$

Let a_x , a_y and a_z be total acceleration ^{components} in x , y and z directions, respectively.

$$a_x = \frac{\delta u}{\delta t} = \frac{\partial u}{\partial x} \frac{\delta x}{\delta t} + \frac{\partial u}{\partial y} \frac{\delta y}{\delta t} + \frac{\partial u}{\partial z} \frac{\delta z}{\delta t} + \frac{\partial u}{\partial t}$$

$$a_x = u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} + \frac{\partial u}{\partial t}$$

$$a_y = \frac{\delta v}{\delta t} = u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} + \frac{\partial v}{\partial t}$$

$$a_z = \frac{\delta w}{\delta t} = u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} + \frac{\partial w}{\partial t}$$

Similarly

Total acceleration

Convective acceleration

i.e.

instantaneous space rate of change of velocity

local acceleration,

i.e.

local time rate change of velocity.

Total acceleration vector $\vec{a}$ is given as

$$\vec{a} = a_x \vec{i} + a_y \vec{j} + a_z \vec{k}$$

Tutorial problem (Solved)

The velocity vector in an incompressible flow is given by

$$\vec{V} = (6xt + yz^2)\mathbf{i} + (3t + xy^2)\mathbf{j} + (xy - 2xyz - 6tz)\mathbf{k}$$

(i) Verify whether the continuity eqn is satisfied?

(ii) Determine the acceleration vector at point A (1,1,1) at $t=1.0$.

Solution

$$u = 6xt + yz^2, \quad v = 3t + xy^2, \quad w = xy - 2xyz - 6tz$$

$$\frac{\partial u}{\partial x} = 6t, \quad \frac{\partial v}{\partial y} = 2xy, \quad \frac{\partial w}{\partial z} = -2xy - 6t$$

(i) from Continuity eqn, $\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 6t + 2xy - 2xy - 6t = 0$
Hence continuity eqn is satisfied.

$$\begin{aligned} (ii) \quad a_x &= \frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} \\ &= 6x + (6xt + yz^2)(6t) + (3t + xy^2)(2xy) + (xy - 2xyz - 6tz)(yz) \\ a_y &= \frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} \\ &= 3 + (6xt + yz^2)(y^2) + (3t + xy^2)(2xy) + (xy - 2xyz - 6tz)(0) \\ a_z &= \frac{\partial w}{\partial t} + u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} \\ &= -6z + (6xt + yz^2)(y - yz) + (3t + xy^2)(x - 2yz) \\ &\quad + (xy - 2xyz - 6tz)(-xy - 6t) \end{aligned}$$

at point (1,1,1) and $t=1.0$

$$a_x = 6 + (6+1)(6) + (3+1)(1) + (1-2-6)^2 = 6 + 42 + 4 - 14 = 38 \text{ units}$$

$$a_y = 3 + (6+1)(1) + (3+1)(2) + (1-2-6)(0) = 18 \text{ units}$$

$$a_z = -6 + (6+1)(1-2) + (3+1)(1-2) + (1-2-6)(-2-6) = 39 \text{ units}$$

$$\vec{a} = \underline{\underline{38\mathbf{i} + 18\mathbf{j} + 39\mathbf{k}}}$$

Tutorial Problem (Solved)

A fluid flow field is given by

$$\vec{V} = x^2y \mathbf{i} + y^2z \mathbf{j} - (2xyz + yz^2) \mathbf{k}$$

Prove that this is a case of possible steady incompressible fluid flow. Calculate the velocity and acceleration at the point (2, 1, 3).

Solution:

$$u = x^2y, \quad v = y^2z, \quad w = -(2xyz + yz^2)$$

$$\frac{\partial u}{\partial x} = 2xy, \quad \frac{\partial v}{\partial y} = yz, \quad \frac{\partial w}{\partial z} = -2xy - yz$$

from Continuity eq. $\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 2xy + yz - 2xy - yz = 0$
i.e. possible case.

$$\vec{V} = (2)^2 \cdot 1 \mathbf{i} + 1^2 \cdot 3 \mathbf{j} - (2 \cdot 2 \cdot 1 \cdot 3 + 1 \cdot 3^2) \mathbf{k}$$

$$= 4\mathbf{i} + 3\mathbf{j} - 21\mathbf{k} \quad \text{ms}$$

velocity magnitude $V = \sqrt{u^2 + v^2 + w^2} = \sqrt{4^2 + 3^2 + (-21)^2} = \sqrt{466} = \underline{\underline{21.587 \text{ ms}}}$

acceleration $a_x = u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} = x^2y(2xy) + y^2z(2^2) - (2xyz + yz^2) \cdot 0$
 $= 2x^3y^2 + x^2y^2z = 2 \cdot 2^3 \cdot 1^2 + 2^2 \cdot 1^2 \cdot 3$
 $= 16 + 12 = \underline{\underline{28}}$

$a_y = u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} = (x^2y) \cdot 0 + y^2z(yz) - (2xyz + yz^2) \cdot 1$
 $= y^3z^2 - 2xy^2z - y^2z^2$
 $= 2 \cdot 1^3 \cdot 3^2 - 2 \cdot 2 \cdot 1^2 \cdot 3 - 1^2 \cdot 3^2$
 $= 18 - 12 - 9 = \underline{\underline{-3}}$

$a_z = u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} = (x^2y)(-yz) + y^2z(-2xz - yz)$
 $+ (-2xyz - yz^2)(-2xy - yz)$

$a = \sqrt{a_x^2 + a_y^2 + a_z^2}$
 $a_z = 1^2 \cdot 3 [2x^2 + 3 + 4x \cdot 2 \cdot 3]$
 $= 3 [8 + 9 + 24]$
 $= 61 \cdot 3$
 $\underline{\underline{183}}$

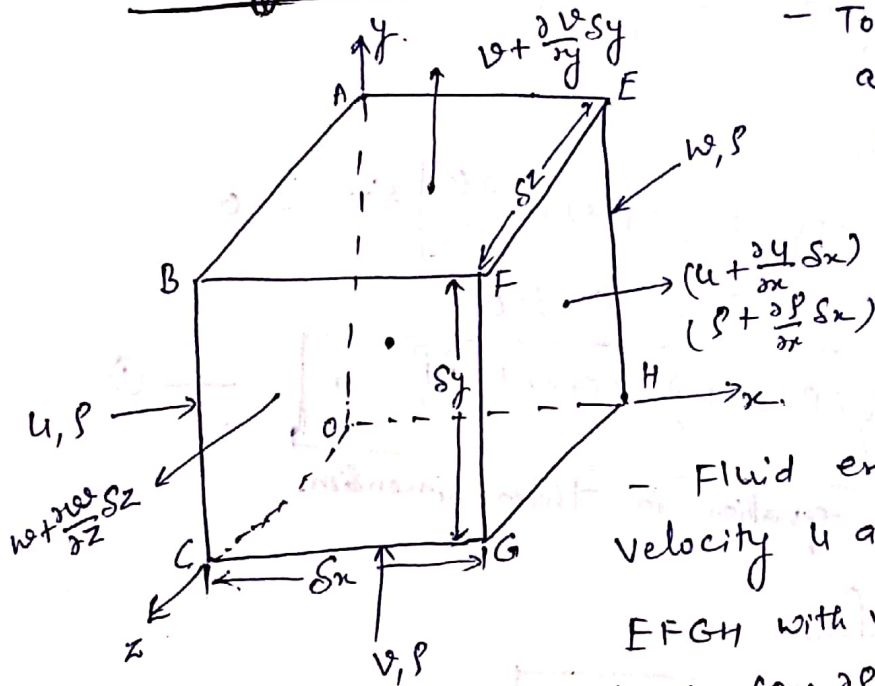
$= -2x^2y^2z - 2xy^2z^2 - y^2z^3$
 $+ 4x^2y^2z + 4xy^2z^2 + 2xy^2z^2 + 2y^2z^3$
 $= y^2z [2x^2 + z^2 + 4xz]$

$a = \sqrt{28^2 + (-3)^2 + 183^2}$
 $= \sqrt{126.18} \text{ ms}^2$

Continuity equation

Continuity equation is derived on the principle of Conservation of mass, according to which in a non nuclear process matter can neither be created nor destroyed. This means mass of fluid per unit time entering at one end of control volume is equal to mass of fluid leaving at other end of control volume plus mass of fluid accumulated in control volume.

Continuity equation in three dimension for Cartesian coordinates



- To derive the Continuity equation at a point in a fluid, a control volume in form of rectangular parallelepiped is considered

- Fluid entering face ABCD with velocity u and density ρ leave face EFGH with velocity $(u + \frac{\partial u}{\partial x} \Delta x)$ and density $(\rho + \frac{\partial \rho}{\partial x} \Delta x)$

$$\text{mass rate entering } ABCD = \rho u \Delta y \Delta z$$

$$\begin{aligned} \text{mass rate leaving } EFGH &= (\rho + \frac{\partial \rho}{\partial x} \Delta x) (u + \frac{\partial u}{\partial x} \Delta x) \Delta y \Delta z \\ &= \rho u \Delta y \Delta z + \rho \frac{\partial u}{\partial x} \Delta x \Delta y \Delta z + u \frac{\partial \rho}{\partial x} \Delta x \Delta y \Delta z \\ &\quad + \text{smaller terms} \end{aligned}$$

$$\begin{aligned} \text{net mass leaving } x\text{-direction} &= \\ &= (\rho \frac{\partial u}{\partial x} + u \frac{\partial \rho}{\partial x}) \Delta x \Delta y \Delta z \\ &= \frac{\partial}{\partial x} (\rho u) \delta V \end{aligned}$$

$$\delta V = \text{Volume of Control Volume}$$

net mass leaving y-direction
 $= \frac{\partial}{\partial y} (\rho v) \delta V$

net mass leaving z-direction
 $= \frac{\partial}{\partial z} (\rho w) \delta V$

Rate of accumulation of mass within control volume is given by

$$\frac{\partial}{\partial t} (\rho \delta x \delta y \delta z) = \frac{\partial \rho}{\partial t} \delta V$$

by mass balance

$$\left[\frac{\partial}{\partial x} (\rho u) + \frac{\partial}{\partial y} (\rho v) + \frac{\partial}{\partial z} (\rho w) + \frac{\partial \rho}{\partial t} \right] \delta V = 0$$

Since $\delta V \neq 0$

$$\boxed{\frac{\partial}{\partial x} (\rho u) + \frac{\partial}{\partial y} (\rho v) + \frac{\partial}{\partial z} (\rho w) + \frac{\partial \rho}{\partial t} = 0} \quad \text{--- ①}$$

This is continuity equation in three dimension.

In vector form.

$$\boxed{\nabla \cdot (\rho \vec{U}) + \frac{\partial \rho}{\partial t} = 0}$$

$$\nabla = \frac{\partial}{\partial x} \vec{i} + \frac{\partial}{\partial y} \vec{j} + \frac{\partial}{\partial z} \vec{k}$$

$$\vec{U} = u \vec{i} + v \vec{j} + w \vec{k}$$

Equation ① is Continuity equation

① In three dimensions for general flow

② For compressible or incompressible fluids, because ρ variation is taken into account.

③ For Steady or unsteady flow because variation with respect to time is taken into account.

④ for uniform or non uniform flow since u, v, w are local variables at any point.

① Steady Compressible flow

Steady $\Rightarrow \frac{\partial \rho}{\partial t} = 0$

$$\frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} = 0$$

② Steady incompressible flow

Incompressible $\Rightarrow \rho = \text{Constant}$

$$\frac{\partial \rho}{\partial x} = \frac{\partial \rho}{\partial y} = \frac{\partial \rho}{\partial z} = \frac{\partial \rho}{\partial t} = 0$$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

③ Two-dimensional flow

$w = 0$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

Existence of flow

If a fluid flow with velocity field $\vec{V} = u\vec{i} + v\vec{j} + w\vec{k}$ were to exist. The velocity component must conform with mass conservation principle i.e. for possible flow the velocity components must obey continuity equation.

Tutorial Problem (Solved)

Two components of velocity in an incompressible ^{fluid} flow are given as

$$u = x^2 - y^2 \quad \text{and} \quad v = z^2 - y^2$$

Determine the third component, assuming that the origin is a stagnation point.

Solution

The Continuity eqⁿ requires that

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

$$(2x^2 - 0) + (0 - 2y^2) + \frac{\partial w}{\partial z} = 0$$

$$\frac{\partial w}{\partial z} = 2y^2 - 2x^2 = 2(y^2 - x^2)$$

$$w = 2(y^2 - x^2)z + f(x, y)$$

↳ Constant of integration, independent of z but may be function of x and y .

origin is a stagnation point

i.e. at $(0, 0, 0)$, $u = v = w = 0$

$$w = 0, \quad \text{at } (0, 0, 0)$$

$$\Rightarrow f(x, y) = 0$$

$$\boxed{w = 2(y^2 - x^2)z}$$

Rotation and Vorticity

"

The Rotation of a fluid particle at a point is given by average angular velocity of two differential linear elements of fluid particle originally perpendicular to each other."

— Consider rectangular element & Rect of fluid. Dotted line shows position of rectangular & Rect after time δt .

— At time t point A has velocities u and v in x and y directions respectively.

— Point D has velocity

$(u + \frac{\partial u}{\partial y} \delta y)$ in x -direction

— Point B has velocity $v + \frac{\partial v}{\partial x} \delta x$ in y -direction.

— Due to difference in velocity of A and D points, linear element AD will have angular velocity.

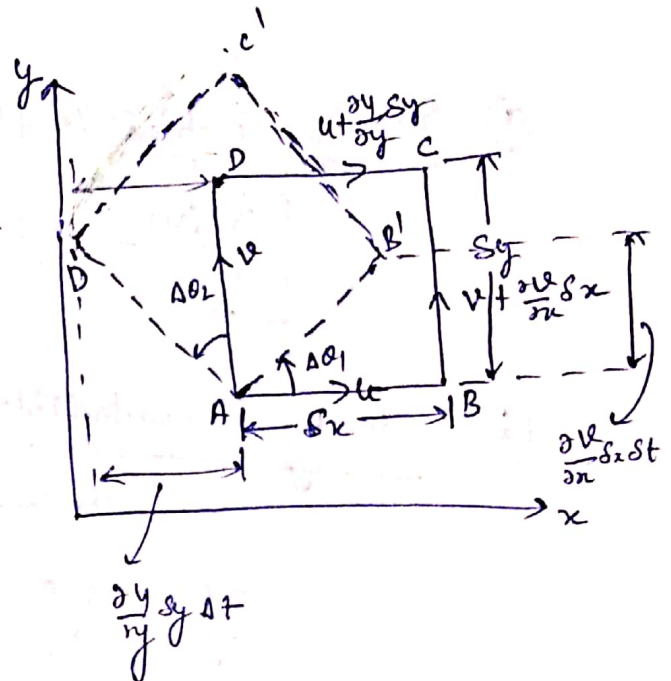
— Similarly due to difference in velocity of A and B points, linear element AB will have angular velocity.

$$\text{Angular velocity of AB} = \frac{\Delta \theta_1}{\delta t} = \frac{\frac{\partial v}{\partial x} \delta x \delta t}{\delta x \delta t} = \frac{\partial v}{\partial x}$$

$$\text{Angular velocity of AD} = \frac{\Delta \theta_2}{\delta t} = \frac{-\frac{\partial u}{\partial y} \delta y \delta t}{\delta y \delta t} = -\frac{\partial u}{\partial y}$$

Rotation about z -axis (motion being in x - y plane)

$$\boxed{\omega_z = \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right)}$$



If the flow is three dimensional, then

Similarly we can write

Rotation about x-axis $w_x = \frac{1}{2} \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \right)$

Rotation about y-axis $w_y = \frac{1}{2} \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right)$

These three scalar equations are put together and written in vector form as

$$\vec{\omega} = i w_x + j w_y + k w_z$$

$$= \frac{1}{2} (\nabla \times \vec{V})$$

$$= \frac{1}{2} \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ u & v & w \end{vmatrix}$$

$$= \frac{1}{2} \text{curl } \vec{V}$$

$$(\vec{V} = u + vj + wk)$$

$$(\nabla \times \vec{V} = \text{curl of velocity vector})$$

Vorticity: Vorticity in a fluid motion is taken numerically equal to twice the value of rotation.

$$\Omega = 2 \vec{\omega} \Rightarrow \vec{\omega} = \frac{\Omega}{2}$$

$$\text{Rotation} = \frac{1}{2} \text{vorticity}$$

For irrotational flow

each component of rotation or vorticity must vanish i.e. $w_x = w_y = w_z = 0$

For 2-D flow in xy plane $w_z = 0 \Rightarrow \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) = 0 \Rightarrow \boxed{\frac{\partial v}{\partial x} = \frac{\partial u}{\partial y}}$

Tutorial problem (solved)

For the following flows, determine the components of rotation about the various axes.

(i) $u = xy^3z$, $v = -y^2z^2$, $w = yz^2 - \frac{y^3z^2}{2}$

(ii) $u = 3xy$, $v = \frac{3}{2}x^2 - \frac{3}{2}y^2$

(iii) $u = y^2$, $v = -3x$

Solution

Components of rotation about various axes

Rotation about Z-axis $\omega_z = \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right)$

- " - x-axis $\omega_x = \frac{1}{2} \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \right)$

- " - y-axis $\omega_y = \frac{1}{2} \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right)$

(i) $\omega_z = \frac{1}{2} (0 - 3xy^2z) = -\frac{3}{2}xy^2z$

$$\omega_x = \frac{1}{2} (z^2 - \frac{3}{2}y^2z^2 + 2y^2z)$$

$$\omega_y = \frac{1}{2} (xy^3 - 0) = \frac{1}{2}xy^3$$

(ii) $\omega_z = \frac{1}{2} (3x - 3x) = 0$

∴ as the flow is having only u and v components.
i.e. flow is two dimensional in xy plane

$$\underline{\underline{\omega_x = \omega_y = 0}}$$

(iii) $\omega_z = \frac{1}{2} (-3 - 2y)$

flow is two dimensional in xy plane

$$\underline{\underline{\omega_x = \omega_y = 0}}$$

Tutorial problem (Solved)

(i) The velocity distribution for a three dimensional flow is

$$u = a + by - cz$$

$$v = d - bx - ez$$

$$w = f + cx - ey$$

where a, b, c, d, e and f are arbitrary constants. Show that these are the velocity components of a fluid motion?

(ii) Does these velocity components represent irrotational flow? If not, determine vorticity and rotation.

Solution

(i) For a possible fluid motion, continuity eqn should be satisfied by velocity components

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

$$0 + 0 + 0 = 0 \quad (\text{satisfied})$$

Hence flow is possible

(ii) For irrotational flow each component of rotation must be zero,

$$\omega_x = \omega_y = \omega_z = 0$$

$$\omega_z = \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) = \frac{1}{2} (-b - b) = -b$$

$$\omega_x = \frac{1}{2} \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \right) = \frac{1}{2} (-e - (-e)) = 0$$

$$\omega_y = \frac{1}{2} \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right) = \frac{1}{2} (-c - c) = -c$$

$$\vec{\omega} = i\omega_x + j\omega_y + k\omega_z = \underline{\underline{-bj - ck}}$$

$$\omega = \sqrt{b^2 + c^2}$$

$$\text{Vorticity} = 2 \cdot \text{rotation}$$

$$\text{Vorticity} = \underline{\underline{2\sqrt{b^2 + c^2}}}$$

Tutorial Problem (Solved)

The velocity components in a two-dimensional flow field for an incompressible fluid are expressed as

$$u = y^2 - x^2 + y$$

$$v = 2xy + x$$

- (i) Show that these functions represent a possible case of an irrotational flow
- (ii) obtain an expression for Stream function ψ .
- (iii) obtain an expression for velocity potential ϕ .

Solution

- (i) For possible flow $\boxed{\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0}$
2D Continuity eqⁿ should be satisfied

$$-2x + 2x = 0$$

(Hence flow is possible)

for irrotational flow

$$\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = 0$$

$$(2y + 1) - (2y + 1) = 0$$

(Hence flow is irrotational)

$$v = - \frac{\partial \psi}{\partial x}$$

$$-(2xy + x) = - \frac{\partial \psi}{\partial x}$$

$$-(x^2y + \frac{x^2}{2}) + g(y) = \psi$$

$$y^2 - x^2 + y = \frac{\partial \psi}{\partial y}$$

$$\frac{y^3}{3} - x^2y + \frac{y^2}{2} + f(x) = \psi$$

inspecting eqⁿ.

$$\boxed{\psi = \frac{y^3}{3} - x^2y + \frac{y^2}{2} + \frac{x^2}{2}}$$

Fluid dynamics

"Dynamics means study of state of motion of a body (fluid mass) under the influence of external forces and moments."

Equation of motion

The dynamics of fluid flow is analysed by Newton's second law of motion which relates acceleration of fluid flow with the forces.

Net force acting on a fluid element is given as

$$\Sigma F = ma$$

Both the forces and acceleration are vector quantities, hence they are resolved in respective directions.

$$\left\{ \begin{array}{l} \Sigma F_x = ma_x \\ \Sigma F_y = ma_y \\ \Sigma F_z = ma_z \end{array} \right\}$$

Forces involved in the fluid motion

In a fluid flow following forces influence the fluid motion

(i) GRAVITY FORCE (F_g)

It is due to weight of fluid and equals mg .

(ii) PRESSURE FORCE (F_p)

Force exerted on fluid mass if there is pressure gradient between two points in the direction of flow.

(iii) VISCOUS FORCE (F_v)

Force due to viscosity of flowing fluid and exists in case of all real fluids.

(iv) TURBULENCE FORCE (F_t)

It is due to turbulence in the flow. It is due to continuous momentum transfer between adjacent layers resulting in development of additional stress (Reynold stress) for flowing fluid.

(v) SURFACE TENSION FORCE (F_s)

It is due to cohesive property of fluid. It is important only when depth of flow is very small.

(vi) COMPRESSIBILITY FORCE (F_c)

It is due to elastic properties of fluid. It is important only for compressible fluids.

Equations of motion

Net force acting on a fluid element is given by

$$\sum F = ma = F_g + F_p + F_v + F_t + F_s + F_c$$

REYNOLD'S EQUATION OF MOTION

In most problems surface tension force and compressibility force are negligible.

$$ma = F_g + F_p + F_v + F_t \rightarrow \text{useful for turbulent flows.}$$

NAVIER STOKES EQUATION OF MOTION

For laminar or viscous flow, turbulent forces are negligible.

$$ma = F_g + F_p + F_v \rightarrow \text{useful for viscous flows.}$$

EULER'S EQUATION OF MOTION

If the fluid is assumed ideal, viscous forces are ignored.

$$ma = F_g + F_p \rightarrow \text{useful for ideal fluid.}$$

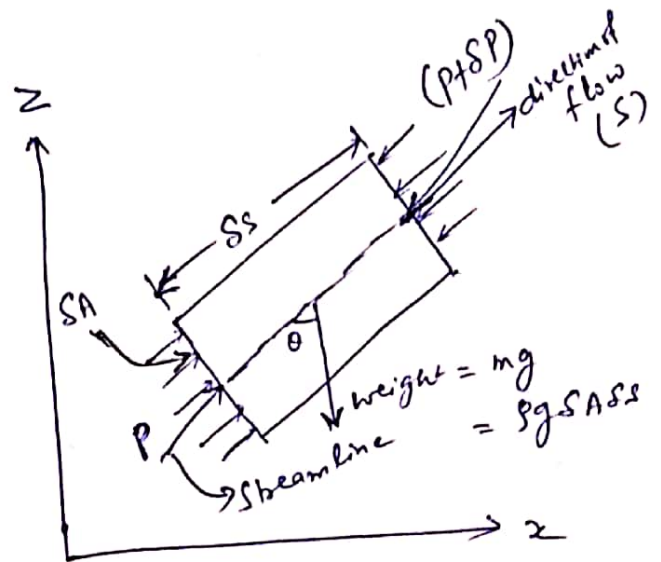
Euler's equation of motion

Consider steady flow of ideal fluid along a stream line.

S = direction of flow along streamline

Consider small element of fluid of cross section area δA and length δs .

θ = angle between direction of flow and line of direction of weight element.



ASSUMPTIONS

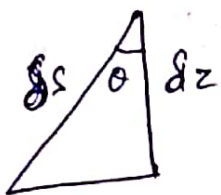
- (i) Fluid is non-viscous.
- (ii) Fluid is incompressible.
- (iii) Flow is steady.
- (iv) Velocity is uniform over cross section.
- (v) Flow is along streamline.
- (vi) Except gravity and pressure forces, no other forces or energy is involved.

Net pressure force acting in the direction of flow

$$P \delta A - (P + \delta P) \delta A = - \delta P \delta A$$

Component of weight of fluid in the direction of flow

$$\begin{aligned}
 &= - \rho g \delta A \delta s \cos \theta \\
 &= - \rho g \delta A \delta s \frac{dz}{\delta s} \\
 &= - \rho g \delta A \delta z
 \end{aligned}$$



Velocity of an elementary fluid particle along a streamline is a function of position and time.

$$v = f(s, t)$$

$$dv = \frac{\partial v}{\partial s} ds + \frac{\partial v}{\partial t} dt$$

$$\text{acceleration} = a_s = \frac{dv}{dt} = \frac{\partial v}{\partial s} \frac{ds}{dt} + \frac{\partial v}{\partial t}$$

$$\boxed{a_s = v \frac{dv}{ds}}$$

for steady flow

$$\frac{\partial v}{\partial t} = 0$$

and partial differentials become total differentials.

From Newton's second law of motion

$$\sum F_s = m a_s$$

$$- \rho \delta A - \rho g \delta A \delta z = \rho (\delta A \delta s) v \frac{dv}{ds}$$

dividing both sides by $-\rho \delta A$

$$+ \frac{\delta p}{\rho} + g \delta z + v dv = 0$$

$$\boxed{\frac{dp}{\rho} + g dz + v dv = 0}$$

→ Euler's equation of motion for steady flow

of an ideal fluid along a streamline.

Bernoulli's equation: Integration of Euler's equation for flow along streamline

If the fluid is incompressible i.e. $\rho = \text{constant}$, Then Euler's equation can be integrated.

$$\int \frac{dp}{\rho} + \int g dz + \int v dv = C$$

$$\frac{p}{\rho} + \frac{v^2}{2} + gz = C$$

Pressure head or static head → Velocity head → datum head

$$\frac{P}{\rho g} + \frac{v^2}{2g} + Z = C$$

This is Bernoulli's equation for steady non-viscous, incompressible fluid along a streamline.

"Hence in steady flow of non-viscous, incompressible fluid the sum of pressure head, velocity head and datum head remains constant at every section."

Restatement

"Sum of pressure energy, kinetic energy and potential energy of a steady ideal fluid flow along streamline is constant."

Some facts

— Bernoulli's equation is derived based on assumptions (stated earlier) and corresponds to limitations of its applications.

— For case of a real fluid flow, some energy of flowing fluid is converted to ^{into} heat due to viscosity and turbulent shear. Hence eqn shows an energy loss.

$$\frac{P_1}{\rho g} + \frac{v_1^2}{2g} + z_1 = \frac{P_2}{\rho g} + \frac{v_2^2}{2g} + z_2 + h_L$$

Flow through a confined passage

— Flow through a pipe, converging or diverging duct, a bend, sudden expansion, sudden contraction, the velocity change is governed by equation of continuity. Consequently due to change in velocity, the pressure change is governed by Bernoulli's equation.

Applications of Bernoulli's equation

- measurement of discharge through a pipe by venturimeter, flow nozzle and orifice meter.
- measurement of static, stagnation and dynamic pressure and hence velocity by pitot-static tube.

measurement of discharge through a pipe by

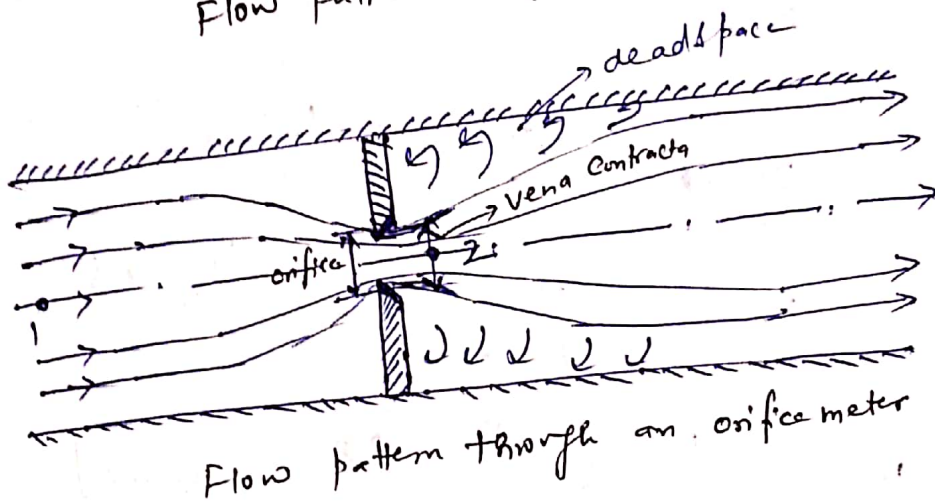
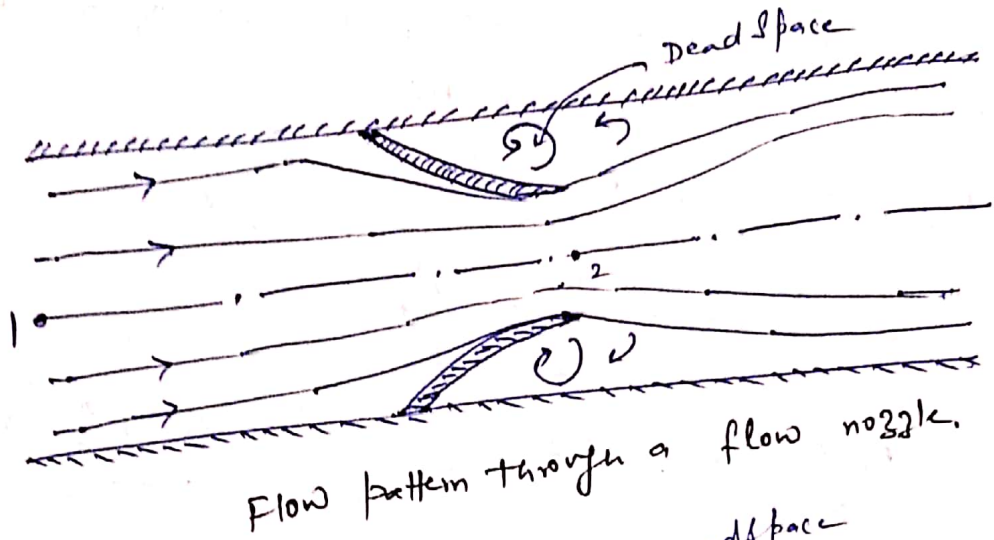
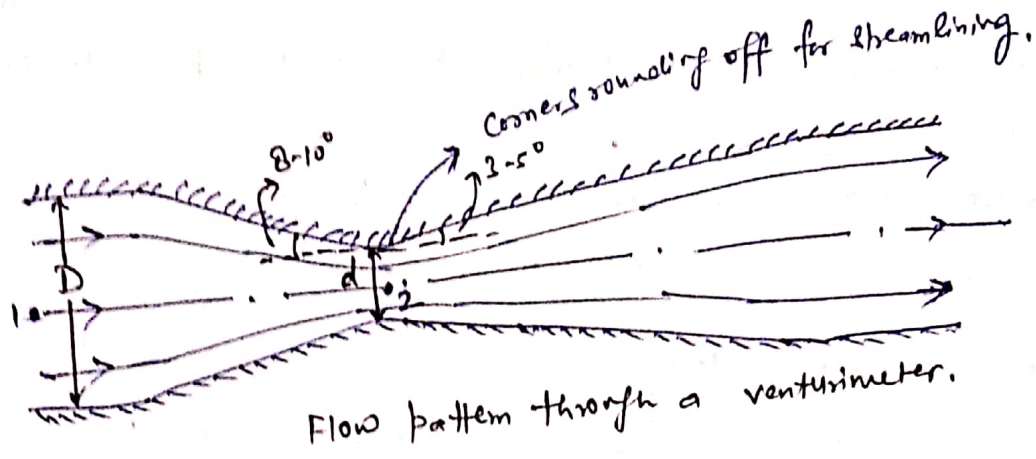
Venturimeter, flow nozzle, orifice meter

"Discharge through a pipe is usually measured by providing a coaxial area contraction within the pipe with a corresponding pressure decrease and causing a pressure difference between two sections. This difference in pressure head is measured by differential manometer and according to Bernoulli's equation is equal to ^{difference in} velocity head between two sections."

Venturimeter

- Venturimeter is a device consisting of short length of gradual convergence, a cylindrical throat and longer length of gradual divergence.
- Semi angle of convergence 20° and semi-angle of divergence is $3-5^{\circ}$.
- pressure tapping 1 is provided before convergence section and pressure tapping 2 is provided at the throat.
- pressure difference between two tappings is measured by means of U-tube manometer with water or mercury as suitable manometric fluid.
- Length of throat is equal to d .

d = diameter of throat.



Flow nozzle It is a device through which the contraction of area is brought about by a streamlined convergent nozzle. Here minimum cross-section of flow exists at the exit of the nozzle. Hence one of the pressure tapping is provided at a distance of one diameter upstream the nozzle plate and the other at the nozzle exit.

Orificemeter An orificemeter is a circular plate with an orifice in its centre. The flow adjusts itself such that it contracts until a point downstream of orifice plate (Vena Contracta) and then expands to fill the passage. one pressure tapping is at one diameter upstream of orifice plate and other at vena contracta (approximately at half the pipe diameter downstream of plate).

- In these devices the 2nd tapping is provided at a point where cross section of fluid is minimum.
- Pressure tapping 1 is free stream inlet to device.
- In venturimeter point 2 is at throat.

Expression for rate of flow

Assuming the flow to be inviscid and incompressible and one dimensional between sections 1 and 2, The Continuity equation and Bernoulli's equation may be applied.

Continuity :
eqn $Q = A_1 V_1 = A_2 V_2 \Rightarrow V_1 = \frac{A_2}{A_1} V_2$

Bernoulli's :
eqn $\frac{P_1}{\rho g} + \frac{V_1^2}{2g} + Z_1 = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + Z_2$

$$\frac{P_1 - P_2}{\rho g} + (Z_1 - Z_2) = \frac{V_2^2 - V_1^2}{2g} = \frac{V_2^2 - \left(\frac{A_2}{A_1}\right)^2 V_2^2}{2g}$$

$$\frac{V_2^2}{2g} \left[1 - \left(\frac{A_2}{A_1}\right)^2 \right] = \frac{P_1 - P_2}{\rho g} + (Z_1 - Z_2)$$

$$V_2^2 = \frac{A_1^2}{A_1^2 - A_2^2} \cdot 2g \left[\frac{P_1 - P_2}{\rho g} + (Z_1 - Z_2) \right]$$

$$V_2 = \frac{A_1}{\sqrt{A_1^2 - A_2^2}} \sqrt{2g \left[\frac{P_1 - P_2}{\rho g} + (Z_1 - Z_2) \right]}$$

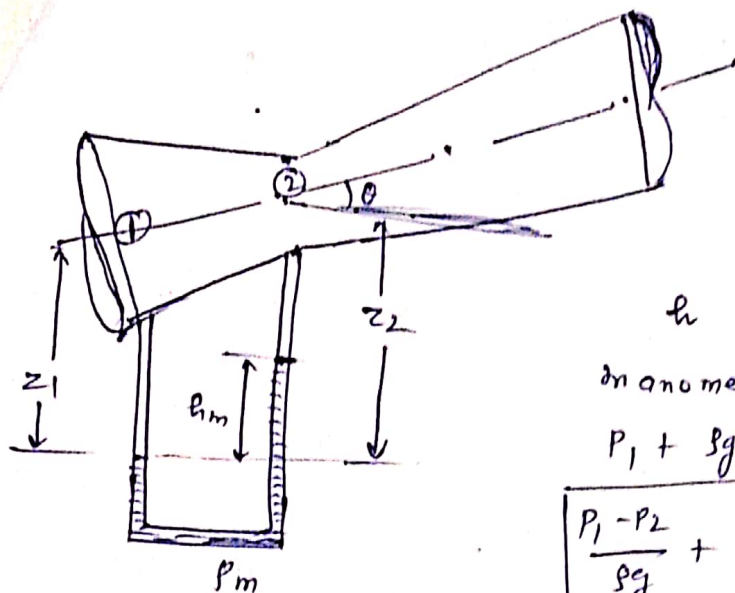
$$Q = A_1 V_1 = A_2 V_2$$

$$Q = \frac{A_1 A_2}{\sqrt{A_1^2 - A_2^2}} \sqrt{2g \left[\frac{P_1 - P_2}{\rho g} + (Z_1 - Z_2) \right]}$$

This is the expression for theoretical discharge.

$$Q = \frac{A_1 A_2}{\sqrt{A_1^2 - A_2^2}} \sqrt{2g h}$$

h = Venturi head = Difference between piezometric heads at sections 1 and 2.



$$\left(\frac{P_1}{\rho g} + z_1 \right) - \left(\frac{P_2}{\rho g} + z_2 \right) = h$$

h can be found out from differential manometer.

$$P_1 + \rho g z_1 - \rho_m g h_m - \rho g (z_2 - h_m) = P_2$$

$$\boxed{\frac{P_1 - P_2}{\rho g} + (z_1 - z_2) = h_m \left(\frac{\rho_m}{\rho} - 1 \right)}$$

- Hence if venturimeter is used to measure discharge in a pipe, which is in inclined position, the value of h may simply be found out by measuring difference in levels of manometric liquid.

Coefficient of discharge

The actual discharge through the flow meter is less than that given by theoretical expression derived on basis of incompressible, inviscid and one dimensional flow.

- In actual practice, flow is not one dimensional.
 - velocity across a cross section does not remain constant. This is due to real flow effects eg. boundary layer formation near the wall and eddy formation in the vicinity of vena contracta (particularly in flow nozzle and orificemeter)
- In orificemeter area of vena contracta is less than the area provided by orifice.
- To take into account wall friction and area contraction effects, a multiplying factor C_d is taken

$$C_d = \text{Coefficient of discharge} = \frac{\text{actual discharge}}{\text{Theoretical discharge}}$$

$$Q_{\text{actual}} = C_d \frac{A_1 A_2}{\sqrt{A_1^2 - A_2^2}} \int 2g h_m \left(\frac{h_m}{s} - 1 \right)$$

Some facts

- In convergent section of venturimeter, the fluid accelerates from inlet section to throat section and then decelerates to fill the passage thereafter. The acceleration is allowed to take place rapidly in small length. However, if retardation was allowed to take place rapidly in small length, the flowing fluid will be separated from boundary walls and eddies are formed which results in excessive energy loss. Hence, divergence cone is made longer. Since separation occurs in divergence cone, this portion is not used for discharge measurement.
- For greater accuracy of pressure difference between inlet and throat section, throat section should be reduced considerably but in the limit of vapour pressure at this section to avoid cavitation phenomenon.
- Generally $d = \frac{1}{2} D$.
- Orificemeter principle is same as of venturimeter but it is a cheaper arrangement. Also where the space is limited, orificemeter may be used for measurement of discharge through pipes. Orificemeter is used for measuring discharge in large diameter pipes.
- Range of C_d

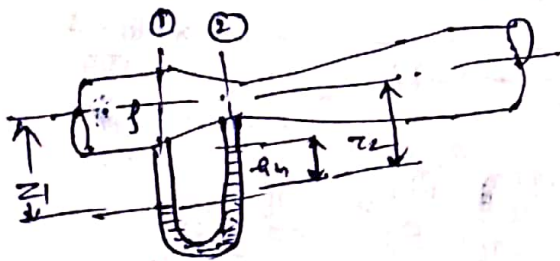
Venturimeter	0.95 - 0.98
Orificemeter	0.6 - 0.65
Flow nozzle	0.7 - 0.8

Tutorial Problem (Solved)

A venturimeter with 200mm diameter at inlet and 100mm throat is laid with axis horizontal and is used for measuring the flow of oil of specific gravity 0.8. The difference of levels in the V-tube differential manometer reads 180 mm of mercury whilst 11.52×10^3 kg of oil is collected in 4 minutes. Calculate coefficient of discharge for venturimeter. Sp. gravity of mercury is 13.6.

Solution: $Q = \frac{A_1 A_2}{\sqrt{A_1^2 - A_2^2}} \sqrt{2gh}$

$h = \text{Venturi head} = \text{difference of Piezometric heads at 1 \& 2}$
 $h = \left(\frac{P_1}{\rho g} + z_1 \right) - \left(\frac{P_2}{\rho g} + z_2 \right) = \frac{P_1 - P_2}{\rho g} + (z_1 - z_2)$



$$P_1 + \rho g z_1 = \rho_m g h_m + \rho g (z_2 + h) = P_2$$

$$\frac{P_1 - P_2}{\rho g} + \frac{z_1 - z_2}{g} = h_m \left(\frac{\rho_m}{\rho} - 1 \right)$$

$$A_1 = \frac{\pi}{4} (0.2)^2 = 0.0314 \text{ m}^2$$

$$A_2 = \frac{\pi}{4} (0.1)^2 = 0.00785 \text{ m}^2$$

$$Q =$$

$$\frac{\frac{\pi}{4} \times (0.2)^2 \times \frac{\pi}{4} \times (0.1)^2}{\left[\left(\frac{\pi}{4} (0.2)^2 \right)^2 - \left(\frac{\pi}{4} (0.1)^2 \right)^2 \right]} \sqrt{2 \times 9.8 \times h_m \left(\frac{\rho_m}{\rho} - 1 \right)}$$

$$= \frac{0.0314 \times 0.00785}{\sqrt{(0.0314)^2 - (0.00785)^2}} \times \sqrt{2 \times 9.8 \times 0.18 \times 13.6}$$

$$\left(\frac{\rho_m}{\rho} - 1 \right)$$

$$= \left(\frac{13.6}{0.8} - 1 \right)$$

$$= 16.25$$

$$= 0.06091 \text{ m}^3 \text{ (theoretical)}$$

$$\text{actual} = \frac{11.52 \times 10^3}{0.8 \times 1000 \times 4 \times 60}$$

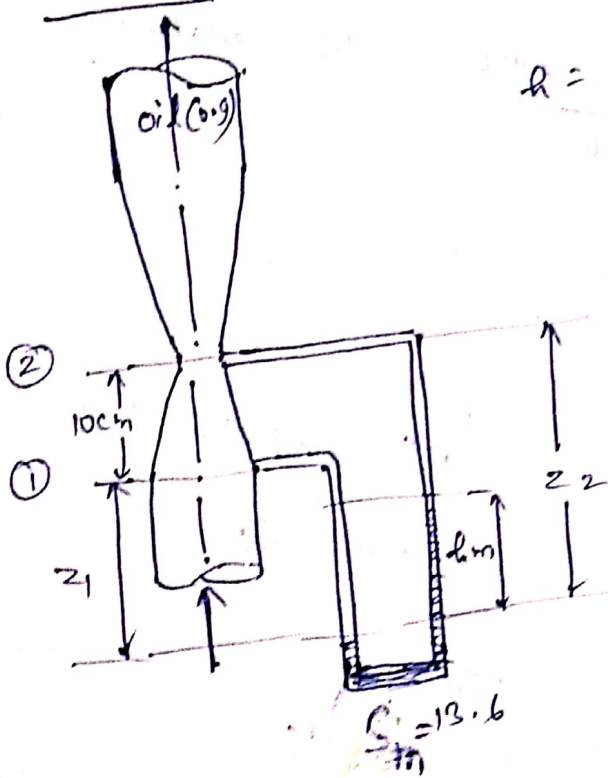
$$= 0.06 \text{ m}^3 \text{ (actual)}$$

$$C_d = \frac{\text{Theoretical discharge}}{\text{actual discharge}} = 0.985$$

Tutorial problem (Solved)

An oil of relative density 0.90 flows through a vertical pipe of diameter 10 cm. The flow is measured by a 20 cm x 10 cm venturimeter. The throat is 10 cm above the inlet section. A differential U-tube manometer containing mercury is connected to the throat and the inlet. If $C_d = 0.99$, what's
 (i) the flow for a manometer reading of 9 cm.
 (ii) the manometer reading for a flow of 50 L/s!

Solution



$$P_1 + \rho g z_1 - \rho_m g h_m - \rho g (z_2 - h_m) = P_2$$

$$h = \frac{P_1 - P_2}{\rho g} + (z_1 - z_2) = h_m \left(\frac{\rho_m}{\rho} - 1 \right)$$

$$Q = \frac{C_d A_1 A_2}{\sqrt{A_1^2 - A_2^2}} \sqrt{2g h}$$

$$A_1 = \frac{\pi}{4} \times (0.2)^2 = 0.0214 \text{ m}^2$$

$$A_2 = \frac{\pi}{4} \times (0.1)^2 = 0.00785 \text{ m}^2$$

$$\sqrt{A_1^2 - A_2^2} = 0.0204$$

$$h = 0.9 \left(\frac{13.6}{0.9} - 1 \right) = 1.27 \text{ m}$$

$$(i) \quad Q = \frac{0.99 \times 0.0214 \times 0.00785}{0.0204} \sqrt{1.27 \times 2 \times 9.8}$$

$$= 0.04 \text{ m}^3/\text{s} = 40 \text{ L/s}$$

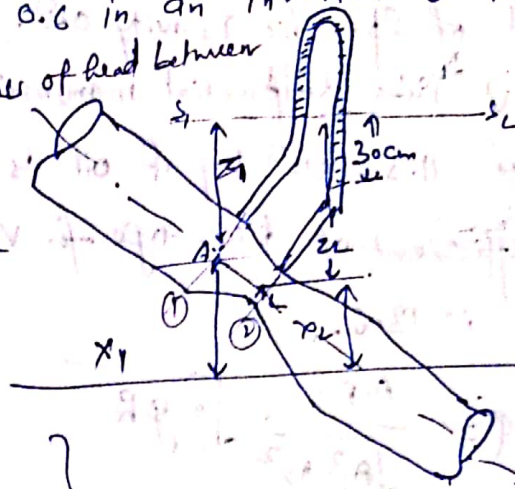
$$(ii) \quad Q = 50 \text{ L/s}$$

$$= 0.05 \text{ m}^3/\text{s}$$

$$\Rightarrow h_m = 0.14 \text{ m} = 14 \text{ cm}$$

Tutorial Problem

Find the discharge of water flowing through a pipe 30 cm diameter placed in an inclined position where a Venturimeter is inserted, having a throat diameter of 15 cm. The difference of pressure between main and throat is measured by a liquid of Sp. Gra. 0.6 in an inverted U-tube which gives a reading of 30 cm. The loss of head between main and throat is 0.2 times the kinetic head of the pipe.



$$Z_1 + x_1 = Z_2 + x_2$$

$$Z_2 - Z_1 = x_1 - x_2$$

Soln.

$$A_1 = \frac{\pi}{4} \times (30)^2$$

$$A_2 = \frac{\pi}{4} \times (15)^2$$

$$P_1 + \rho g Z_1 + \rho g (30) + \rho g (Z_2 - Z_1) = P_2$$

$$(P_1 - P_2) + \rho g (Z_2 - Z_1) = 1000 \times 9.81 \times 30$$

$$= 0.6 \times 9.81 \times 30 \times 1000$$

$$\frac{P_1 - P_2}{\rho g} + (Z_2 - Z_1) = \frac{2(P_1 - P_2)}{\rho g} = 0.3 \left(\frac{P_1 - P_2}{\rho g} \right)$$

$$= 0.3 (1.5) = 0.45$$

$$= 0.12$$

$$= \frac{1177.20}{9.81} = 0.12$$

Bernoulli's

$$\frac{P_1}{\rho g} + \frac{v_1^2}{2g} + x_1 = \frac{P_2}{\rho g} + \frac{v_2^2}{2g} + x_2 + \text{head loss}$$

$$\frac{P_1 - P_2}{\rho g} + (x_1 - x_2) = \frac{v_2^2 - v_1^2}{2g} + 0.2 \times \frac{v_1^2}{2g}$$

$$\Rightarrow 0.12 = \frac{15.2 v_1^2}{2g}$$

Continuity

$$A_1 v_1 = A_2 v_2$$

$$v_2 = v_1 \times \frac{A_1}{A_2} = v_1 \times \frac{(30)^2}{(15)^2} = 4 v_1$$

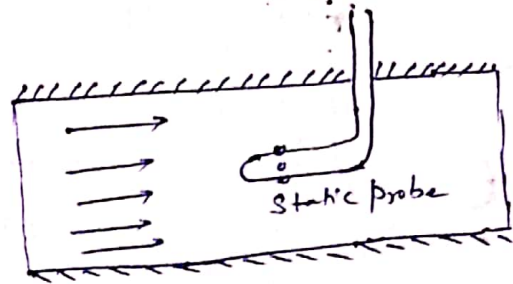
$$v_1 = 39.35 \text{ cm/s}$$

$$Q = A_1 v_1 = \frac{\pi}{4} \times (30)^2 \times (39.35)$$

$$= 0.02782 \text{ m}^3/\text{s}$$

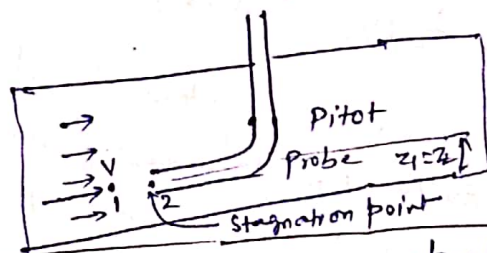
Measurement of static, stagnation and dynamic pressure and velocity

Static pressure in the fluid is measured by a static probe inserted in fluid. The probe has static pressure holes such that the axis of holes is normal to velocity of fluid. There is normally three or more holes on the periphery of the probe and when connected to manometer provide static pressure at the cross section.



STAGNATION PRESSURE at a point in a fluid flow is the total pressure which would result if the fluid were brought to rest isentropically.

A simple device to measure stagnation pressure is a tube with a hole in front inserted in the flow such that velocity is normal to the plane of hole i.e. along the axis of hole. Stagnation point is created by bringing fluid to rest at the nose of the tube and pressure at that point corresponds to stagnation pressure.



$$\boxed{\text{Stagnation pressure} = \text{Static pressure} + \text{dynamic pressure}}$$

Point 1 is taken at which the velocity is V . Point 2 lies immediately at the nose of the tube. This is stagnation point where velocity of flow is reduced to zero.

Applying Bernoulli equation between 1 and 2

$$\frac{P_1}{\rho g} + \frac{V_1^2}{2g} + \frac{z_1}{g} = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + \frac{z_2}{g}$$

Since $z_1 = z_2$ and $V_2 = 0$, we get:

$$P_2 = \underbrace{\frac{P_1}{\rho}}_{\text{Static pressure}} + \underbrace{\frac{1}{2} \rho V_1^2}_{\text{dynamic pressure}}$$

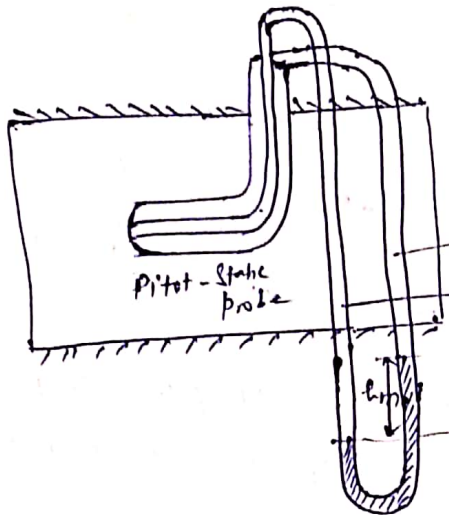
Stagnation pressure

Pitot-static probe measures the difference of the two pressures i.e. stagnation pressure and static pressure using U-tube differential manometer and used to measure velocity of the flow.

$$P_2 = P_1 + \frac{1}{2} \rho V^2$$

$$\frac{V^2}{2g} = \frac{P_2 - P_1}{\rho g} = h_m \left(\frac{\rho_m}{\rho} - 1 \right)$$

$$V = \sqrt{2g h_m \left(\frac{\rho_m}{\rho} - 1 \right)}$$



The whole of the stream impinging the tip may not be brought to rest, so a factor known as coefficient of tube is used to account for this aspect.

$$V = C \sqrt{2g h_m \left(\frac{\rho_m}{\rho} - 1 \right)}$$

$$C = 0.95-1 \text{ (for well designed pitot tube)}$$

Tutorial problem (Solved)

A pipe carries a flow of oil of R.D. = 0.85. A pitot static tube is inserted into the pipe to measure velocity at a point. If the differential mercury oil gauge connected to the pitot-static tube indicates a reading of 4cm, calculate the velocity at that point. Assume the coefficient of pitot tube as 0.99.

Solution

$$V = C \sqrt{2g h_m \left(\frac{\rho_m}{\rho} - 1 \right)}$$

$$= 0.99 \sqrt{2 \times 9.81 \times 0.04 \times \left(\frac{12.6}{0.85} - 1 \right)}$$

$$= \underline{\underline{3.397 \text{ m/s}}}$$

Tutorial problem (Solved)

A pitot tube is inserted in a pipe of 30 cm diameter. The static pressure of the tube is 10 cm of mercury vacuum. The stagnation pressure at the centre of the pipe recorded by pitot tube is 1.0 N/cm^2 . Calculate rate of flow of water through the pipe, if the mean velocity of the flow is 0.85 times the centre line velocity. Assume coefficient of tube to be 0.98.

Solution

$$\frac{P_{\text{stagnation}}}{\rho g} = \frac{1}{10^{-4} \times 1000 \times 9.81} = 1.019 \text{ m of water.}$$

$$\frac{P_{\text{static}}}{\rho g} = \frac{-10 \times 10^{-2} \times 13.6}{1000 \times 9.81} = -1.36 \text{ m of water}$$

$$\frac{P_{\text{stg}} - P_{\text{static}}}{\rho g} = \frac{1.019 - (-1.36)}{1000 \times 9.81} = 2.379 \text{ m}$$

$$V = \sqrt{2g \left(\frac{P_2 - P_1}{\rho g} \right)} = 0.98 \sqrt{2 \times 9.81 \times 2.379} = 6.695 \text{ m/s}$$

$$\text{mean velocity} = 0.85 \times 6.695 = 5.69 \text{ m/s}$$

$$Q = \frac{\pi}{4} d^2 V_{\text{mean}}$$
$$= \frac{\pi}{4} \times (0.3)^2 \times 5.69$$
$$Q = 0.402 \text{ m}^3/\text{s}$$

Dimensional analysis

Introduction Dimensional analysis deals with a process in which all important variables involved in a physical phenomenon are systematically organised into dimensionless groups.

Uses

- 1) Testing dimensional homogeneity of any equation of fluid motion.
- 2) Deriving equations expressed in terms of non-dimensional parameters to show relative significance of each parameter.
- 3) Planning model tests and presenting experimental results in terms of non-dimensional parameters.

Dimensions

All physical quantities are expressed by magnitude and units.

In SI system FUNDAMENTAL or PRIMARY PHYSICAL QUANTITIES which are assigned base units are.

- ✓ 1) Mass
- ✓ 2) Length
- ✓ 3) Time
- ✓ 4) Temperature
- 5) Current
- 6) Luminous Intensity

In fluid mechanics and heat transfer first four are useful and given dimensions as

M, L, T, θ respectively.

DERIVED or SECONDARY QUANTITIES

They are expressed in terms of fundamental quantities.

Note: Sometimes Force (F) is used as fundamental quantity instead of mass. $F-L-T-\theta$ System.

② Dimensions of derivatives and Integral

$$i) \frac{dy}{dx} \Rightarrow \lim_{\Delta x \rightarrow 0} \left(\frac{\Delta y}{\Delta x} \right) = \frac{y}{x}$$

$$ii) \frac{d^2y}{dx^2} \Rightarrow \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d \left(\frac{dy}{dx} \right)}{\Delta x \rightarrow 0} = \frac{\frac{dy}{dx}}{x} = \frac{y/x}{x} = \frac{y}{x^2}$$

$$iii) \int y dx = y \cdot x$$

$$iv) \iint p \cdot dx dy = p \cdot x \cdot y$$

Dimensions of some important physical quantities

- 1) Angular speed (Rad./sec.) = T^{-1}
- 2) Angular acceleration (Rad./sec.²) = T^{-2}
- 3) Angular momentum (Mxrxv) = $M L^2 T^{-1}$
- 4) moment or torque (Fxr) = $M L^2 T^{-2}$
- 5) Stream function, ψ ($u = \frac{\partial \psi}{\partial y}$) = $L^2 T^{-1}$
- 6) velocity potential, ϕ ($u = \frac{\partial \phi}{\partial x}$) = $L^2 T^{-1}$
- 7) Circulation, Γ ($\Gamma = \oint \vec{v} \cdot d\vec{s}$) = T^{-1}
- 8) Vorticity, $\left(\frac{\Gamma}{\Delta A} \right)$ = T^{-1}
- 9) Rotation, $\frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right)$ = T^{-1}
- 10) Surface tension (F/l) = $M T^{-2}$
- 11) Dynamic viscosity (μ) ($\tau = \mu \frac{du}{dy}$) = $M L^{-1} T^{-1}$
- 12) Kinematic viscosity (ν) = μ/ρ = $L^2 T^{-1}$
- 13) modulus of elasticity (E or k) = $M L^{-1} T^{-2}$

Note:

$$\text{Rotation} = \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) = T^{-1}$$

$$\text{Circulation} = \Gamma = \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) dx dy = L^2 T^{-1}$$

$$\text{Vorticity} = \frac{\text{Circulation}}{\text{Area}} = T^{-1}$$

Dimensional homogeneity

Principle of dimensional homogeneity states that "Equations describing a physical phenomenon must be dimensionally homogeneous i.e. the dimensions of various terms on two sides of equations are identical."

example

The equation (Euler's equation of motion)

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = -\frac{1}{\rho} \frac{\partial p}{\partial x}$$

is dimensionally correct.

$$\begin{aligned} [L T^{-1}] \left[\frac{L T^{-1}}{L} \right] + [L T^{-1}] \left[\frac{L T^{-1}}{L} \right] &= [M^{-1} L^3] \left[\frac{M L^{-1} T^{-2}}{L} \right] \\ [L T^{-2}] + [L T^{-2}] &= [L T^{-2}] \end{aligned}$$

dimensions of both sides are same.

Note: There are several empirical relations in which some coefficient is there. If coefficient is assumed dimensionless then the equation is dimensionally non-homogeneous

e.g. Chezy's formula

$$V = C \sqrt{m i}$$

V = velocity through a channel
 m = Hydraulic mean depth or hydraulic radius
 i = slope of channel
 C = Chezy's coefficient.

$$L T^{-1} = 1 [L^{1/2}] [1]$$

(dimensions of both sides are not same)

Hence if equation has to be dimensionally homogeneous then Chezy's coefficient (C) must have dimensions

$$C = \frac{L T^{-1}}{L^{1/2}} = L^{1/2} T^{-1}$$

$$C = L^{1/2} T^{-1}$$

Also Manning coefficient (n) in the Manning formula for velocity of flow in open channel

$$V = \frac{1}{n} m^{2/3} i^{1/4}$$

$$n = L^{1/3} T$$

(4)

Applications of dimensional homogeneity

Principle of dimensional homogeneity serves the following purpose.

- 1) Helps to check whether an equation of any physical phenomenon is dimensionally homogeneous or not.
- 2) It helps to determine the dimensions of a physical quantity.

eg. $q = -KA \frac{dT}{dx}$, we can find the dimensions of thermal conductivity K .

- 3) Dimensional homogeneity helps to convert the units from one system to another.

Dimensional analysis: Buckingham's $\Pi(C\pi)$ theorem

"If there are n variables in a dimensionally homogeneous equation and if these variables contain m primary dimensions. Then these variables can be grouped in $(n-m)$ independent non-dimensional parameters and called $\Pi(C\pi)$ terms."

Note: There can be formed in finite number of dimensionless groups but only $(n-m)$ will be independent.

Procedure for Buckingham method

- ① List all important variables involved in a physical phenomenon.
- ② Note the primary dimensions involved in them and find out the value of m . Then there will be $(n-m)$ non-dimensional groups.
- ③ Now we recognise m repeating variables which are common to all groups desired to be formed. They are so chosen that all m fundamental quantities are included.
- ④ Non repeating variables are taken one by one together with repeating variables.
- ⑤ In fluid flow problem, generally characteristic length (l), characteristic velocity (V) and fluid density (ρ) are taken as repeating variables. The reason is that l , V and ρ are representative of inertial behaviour of the fluid. They include all fundamental dimensions.
 $l [L]$, $V [LT^{-1}]$, $\rho [ML^{-3}]$

Guidelines for recognition of important variables in a problem

- These variables are most important physical quantities in the problem.
 - Non repeating independent variables should be listed as far as possible.
 - These should be geometric (e.g. length), Kinematic (e.g. velocity) and dynamic (e.g. force) variables in the list.
 - Relevant fluid property (e.g. ρ or μ) must be included.
- For heat transfer problem, a thermal property (e.g. thermal conductivity, k) are also chosen.

⑥

- ⊗ None of repeating variable should be dimensionless.
- ⊗ No two variables have same dimensions.
- ⊗ They themselves do not form a dimensionless parameter.
- ⊗ None of repeating variable should be dependent.

⑥ π terms are formed by m repeating variables and one non repeating variable.

eg. $\left\{ \begin{array}{l} \text{viscosity } (\mu) \\ \text{Gravitational acceleration } (g) \\ \text{Surface tension } (\sigma) \end{array} \right\}$
non repeating variables.

⑦ Write expressions for π terms. Raise unknown exponent to repeating variable and put exponent (1) on non-repeating variable.

⑧ Find values of unknown exponents by use of fundamental dimensions.

Some facts about π terms

In order to get expression in desired form, the following points are noted.

- ① If a quantity is dimensionless, it is a π term.
- ② If two quantities have same dimensions, their ratio will be a π term e.g. (H/d) .
- ③ Any π term may be replaced by any power of that term
eg. $\pi_1 \rightarrow \pi_1^{-1}$, $\pi_2 \rightarrow \pi_2^2$, $\pi_3 \rightarrow \frac{1}{\sqrt{\pi_3}}$
- ④ Any π term may be replaced by multiplying it by a numerical constant. eg. $\pi_1 \rightarrow 2\pi_1$
- ⑤ Any π term may be replaced by multiplying or dividing it by another π term
eg. $\pi_1 \text{ or } \pi_2 \rightarrow (\pi_1 \times \pi_2) \text{ or } (\pi_1 / \pi_2)$

Important force ratios: Dimensionless numbers

(7)

(i) Reynolds Number (Re)

It is the ratio of inertia force to the viscous force.

$$Re = \frac{\text{Inertia force}}{\text{Viscous force}}$$

$$\begin{aligned} \text{Inertia force} &= \text{mass} \times \text{acceleration} = (\rho \times \text{Volume}) \times \frac{\text{Velocity}}{\text{time}} \\ &= \rho \times A \times \text{length} \times \frac{\text{Velocity}}{\text{time}} \end{aligned}$$

$$\text{Inertia force} = \rho l^2 v^2$$

$$\text{Viscous force} = \tau \times A = \mu \frac{v}{l} l^2 = \mu v l$$

$$\tau = \mu \frac{dv}{dy}$$

$$Re = \frac{\rho l v^2}{\mu v} = \frac{\rho v l}{\mu}$$

$$Re = \frac{\rho v l}{\mu}$$

l = characteristic length
e.g. diameter of pipe.

Reynolds number will be considered for situations in which viscous force predominates over other forces e.g. gravity force and elastic forces.
e.g. in following situations.

(i) Completely enclosed flow

- ① - incompressible flow through pipes, Venturimeters, orifice meters and flow between parallel plates.

(ii) Completely submerged flow

- ② - Low velocity flow around aeroplanes and submarines
- ③ - Flow around structures and bodies.
- ④ - Flow through low speed aerodynamic machines.

⑧

(ii) Froude number (Fr)

$$Fr = \left(\frac{\text{Inertia force}}{\text{Gravity force}} \right)^{1/2}$$

$$\text{Gravity force} = mg = \rho l^3 g$$

$$Fr = \frac{\sqrt{\rho l^2 v^2}}{\sqrt{\rho l^3 g}} = \frac{v}{\sqrt{lg}}$$

Froude number is used for dynamic similarity when the forces other than gravitational forces are negligible e.g. in

Note! - It is the ratio of mean velocity of ship to velocity of small wave in quiet fluid.
- Its value is considered as critical value.

- (1) Flow in open channels (free surface flows)
- (2) Flow of liquid jets from orifice
- (3) Surface wave motion created by boats & ships.
- (4) Flow over notches, weirs etc.
- (5) problems in which fluid of different mass densities flow over one another.

(iii) Euler Number (Eu)

It is the ratio of pressure force to the inertia force.

$$Eu = \frac{\text{Pressure force}}{\text{Inertia force}} = \frac{\rho \Delta^2}{\rho l^2 v^2}$$

$$\text{Pressure force} = \text{Pressure} \times \Delta^2 = p \Delta^2$$

$$E = \frac{p}{\rho v^2}$$

↓
It is also called pressure coefficient.

Euler number is used when

pressure changes in a flow are predominant.

- (1) Flow through pipes in which turbulence is fully developed and viscous forces are negligible.

- (2) Flow through pen stock leading water to hydraulic turbine.
- (3) Flow over submerged bodies.
- (4) Pressure rise due to sudden closure of valves.
- (5) Also used where phenomenon of cavitation takes place.

(iv) Mach Number (M) $M = \left(\frac{\text{Inertia force}}{\text{Elastic force}} \right)^{1/2}$

Elastic force = bulk modulus of elasticity $\times \text{Area} = K l^2$
 $= \rho c^2 l^2$

$$c = \sqrt{K/\rho}$$

$$\Rightarrow K = \rho c^2$$

c = Speed of sound in flowing medium

$$M = \left(\frac{\rho l^2 v^2}{\rho l^2 c^2} \right)^{1/2}$$

$$M = v/c = \frac{\text{Speed of fluid}}{\text{Speed of sound in that fluid}}$$

Mach number effects are considered only when M exceeds 0.4 and effects of compressibility comes into picture e.g. in

- (i) Aerodynamic testing.
- (ii) Flow of aeroplanes and projectiles through air at supersonic speed.
- (iii) In case unsteady flows with water hammer problems.
- (iv) Flow of gases exceeding velocity of sound.

Transonic flow	$0.9 < M < 1.1$
Supersonic flow	$1 < M < 7$
Hypersonic flow	$M > 7$

(v) Weber Number (W)

$$W = \frac{\text{Inertia force}}{\text{Surface tension force}}$$

$$= \frac{\rho l^2 v^2}{\sigma \times l} = \frac{\rho l v^2}{\sigma}$$

$$\sigma = \frac{\text{Surface tension}}{\text{per unit length}}$$

Weber number is important for following cases.

- (1) Capillary tube flows (Capillary rise in narrow passages, capillary movement of water in soil)
- (2) Human blood flow studies in veins and arteries.
- (3) Thin sheet flows (very thin sheet of fluid flowing over a surface).

Tutorial problem (Solved)

Consider force F on the propeller of an aircraft. It depends on forward speed of aircraft U , density of air ρ , viscosity of air μ , diameter of propeller D and speed of rotation of propeller N . $F = f[U, \rho, \mu, D, N]$
Obtain expression for F in non-dimensional terms.

Solution:

total variables, $n = 6$

Total primary dimensions, $m = 3$, using Buckingham
theorem number of non-dimensional π terms = $n - m = 6 - 3 = 3$

choose repeating variables, D, U, ρ

π_1 will have

π_2 "

π_3 "

$$\left. \begin{array}{l} D U \rho F \\ D U \rho \mu \\ D U \rho N \end{array} \right\} \Rightarrow \begin{array}{l} \pi_1 = D^{a_1} U^{b_1} \rho^{c_1} F \\ \pi_2 = D^{a_2} U^{b_2} \rho^{c_2} \mu \\ \pi_3 = D^{a_3} U^{b_3} \rho^{c_3} N \end{array}$$

① π_1

$$M^0 L^0 T^0 = [L]^{a_1} [L T^{-1}]^{b_1} [M L^{-3}]^{c_1} [M L T^{-2}]$$

$$M^0 L^0 T^0 = [M]^{c_1+1} [L]^{a_1+b_1-3c_1+1} [T]^{-b_1-2}$$

$$c_1+1=0 \Rightarrow c_1=-1, \quad -b_1-2=0 \Rightarrow b_1=-2, \quad a_1+b_1-3c_1+1=0$$

$$a_1-2+3+1=0 \Rightarrow a_1=-2$$

$$\pi_1 = \frac{F}{\rho U^2 D^2}$$

② π_2

$$M^0 L^0 T^0 = [L]^{a_2} [L T^{-1}]^{b_2} [M L^{-3}]^{c_2} [M L^{-1} T^{-1}]$$

$$= [M]^{c_2+1} [L]^{a_2+b_2-3c_2+1} [T]^{-b_2-1}$$

$$c_2+1=0 \Rightarrow c_2=-1, \quad -b_2-1=0 \Rightarrow b_2=-1, \quad a_2+b_2-3c_2+1=0$$

$$a_2-1+3+1=0 \Rightarrow a_2=-1$$

$$\pi_2 = \frac{\mu}{\rho U D}, \text{ replace } \pi_2 \rightarrow \pi_2^{-1}, \quad \pi_2 = \frac{\rho U D}{\mu}$$

③ π_3

$$M^0 L^0 T^0 = [L]^{a_3} [L T^{-1}]^{b_3} [M L^{-3}]^{c_3} [T^{-1}]$$

$$= [L]^{a_3+b_3-3c_3} [M]^{c_3} [T]^{-b_3-1}$$

$$c_3=0, \quad a_3+b_3-3c_3=0 \Rightarrow a_3+b_3=0$$

$$-b_3-1=0 \Rightarrow b_3=-1 \Rightarrow a_3=1$$

$$\pi_3 = \frac{N D}{U}$$

$$\frac{F}{\rho U^2 D^2} = f \left[\frac{\rho U D}{\mu}, \frac{N D}{U} \right]$$

Tutorial problem (Solved) The pressure difference (ΔP) in a pipe of diameter D and length l due to turbulent flow depends on velocity V , viscosity μ , density ρ and average height k of roughness projections. Obtain expression for ΔP . $\Delta P = f [D, l, V, \rho, \mu, k]$

Solution $n = 7, m = 3, n - m = 4$

① π_1

$$m^0 L^0 T^0 = [L]^{a_1} [L T^{-1}]^{b_1} [M L^{-3}]^{c_1} M^{d_1} T^{e_1}$$

$$\begin{bmatrix} \pi_1 - D^{a_1} V^{b_1} \rho^{c_1} \Delta P^{d_1} \\ \pi_2 - D^{a_2} V^{b_2} \rho^{c_2} l^{d_2} \\ \pi_3 - D^{a_3} V^{b_3} \rho^{c_3} k^{d_3} \\ \pi_4 - D^{a_4} V^{b_4} \rho^{c_4} \mu^{d_4} \end{bmatrix}$$

$$a_1 + b_1 - 3c_1 - d_1 = 0 \Rightarrow a_1 - 2 + 3 - 1 = 0$$

$$a_1 = 0$$

$$c_1 + 1 = 0 \Rightarrow c_1 = -1$$

$$-b_1 - 2 = 0 \Rightarrow b_1 = -2$$

$$\pi_1 = D^0 V^{-2} \rho^{-1} \Delta P$$

$$\boxed{\pi_1 = \frac{\Delta P}{\rho V^2}}$$

② π_2

$$m^0 L^0 T^0 = [L]^{a_2} [L T^{-1}]^{b_2} [M L^{-3}]^{c_2} L^{d_2}$$

$$a_2 + b_2 - 3c_2 + d_2 = 0$$

$$a_2 + 1 - 3 \times 0 + 1 = 0 \Rightarrow a_2 = -1$$

$$c_2 = 0$$

$$-b_2 = 0$$

$$\boxed{\pi_2 = \frac{l}{D}}$$

$$\boxed{\pi_3 = k/D}$$

③ π_3

Similarly.

④ π_4

$$m^0 L^0 T^0 = [L]^{a_4} [L T^{-1}]^{b_4} [M L^{-3}]^{c_4} M^{d_4} T^{e_4}$$

$$a_4 + b_4 - 3c_4 - d_4 = 0$$

$$a_4 - 1 + 3 - 1 = 0 \Rightarrow a_4 = -1$$

$$c_4 + 1 = 0, -b_4 + 1 = 0$$

$$c_4 = -1, b_4 = 1$$

$$\boxed{\pi_4 = \frac{\mu}{\rho V D}}$$

$$\frac{\Delta P}{\rho V^2} = f \left[\frac{l}{D}, \frac{k}{D}, \frac{\mu}{\rho V D} \right]$$

$$\boxed{\frac{\Delta P}{\rho V^2} = f \left[\frac{l}{D}, \frac{k}{D}, \frac{\rho V D}{\mu} \right]}$$

It can be written as

Reynolds number.

Tutorial problem (Solved)

Show that velocity through a circular orifice is given by

$$V = \sqrt{2gH} \phi \left[\frac{D}{H}, \frac{\mu}{\rho V H} \right]$$

Solution:

$$V = f [H, D, \mu, \rho, g]$$

$$\begin{aligned} n &= 6 \\ m &= 2 \\ n - m &= 4 \end{aligned}$$

$$\pi_1 \rightarrow M^0 L^0 T^0 = H^{a_1} V^{b_1} \rho^{c_1} D$$

$$\pi_2 \rightarrow M^0 L^0 T^0 = H^{a_2} V^{b_2} \rho^{c_2} g$$

$$\pi_3 \rightarrow M^0 L^0 T^0 = H^{a_3} V^{b_3} \rho^{c_3} \mu$$

$$\pi_1 \rightarrow M^0 L^0 T^0 = L^{a_1} [LT^{-1}]^{b_1} [ML^{-3}]^{c_1} L$$

$$\begin{aligned} a_1 + b_1 - 3c_1 + 1 &= 0, & -b_1 &= 0 \\ a_1 + 0 + 0 + 1 &= 0, & c_1 &= 0 \\ a_1 &= -1 \end{aligned}$$

$$\pi_1 = D/H$$

$$\pi_2 \rightarrow M^0 L^0 T^0 = L^{a_2} [LT^{-1}]^{b_2} [ML^{-3}]^{c_2} LT^{-2}$$

$$\begin{aligned} a_2 + b_2 - 3c_2 + 1 &= 0, & -b_2 - 2 &= 0, & c_2 &= 0 \\ a_2 + 0 + 0 + 1 &= 0, & b_2 &= -2 \\ a_2 &= -1 \end{aligned}$$

$$\pi_2 = \frac{H g}{V^2}$$

$$\pi_3 \rightarrow M^0 L^0 T^0 = L^{a_3} [LT^{-1}]^{b_3} [ML^{-3}]^{c_3} ML^{-1}T^{-1}$$

$$\begin{aligned} c_3 + 1 &= 0, & -b_3 - 1 &= 0, & a_3 + b_3 - 3c_3 - 1 &= 0 \\ c_3 &= -1, & b_3 &= -1, & a_3 - 1 + 3 - 1 &= 0 \Rightarrow a_3 = -1 \end{aligned}$$

$$\pi_3 = \frac{\mu}{\rho V H}$$

replace $\pi_2 = \frac{1}{\sqrt{\pi_2}} = \frac{V}{\sqrt{2gH}}$ replace by $\frac{V}{\sqrt{2gH}}$

$$V = \sqrt{2gH} \phi \left[\frac{D}{H}, \frac{\mu}{\rho V H} \right]$$